Math 104: Introduction to Analysis

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Though it may seem (or not!) as though I put care into these notes, they are actually very sloppily written. So I guarantee you that there will be errors, typos, flat out lies, and other varieties of mistakes. You may alert me if you find one, but the best thing to do is probably to read it with a skeptical and also not-so-literal eye. You should be especially skeptical about anything I say that seems like it has to do with mathematical logic, since I literally know nothing about that and am just making stuff up. Much of the content of these notes is taken from Walter Rudin's *Principles of Mathematical Analysis* and to a lesser extent Kenneth Ross' *Elementary Analysis*, though of course the errors are all mine.

1 Lecture 1

Remark 1.1 (Proof by contradiction)**.** This is something you should become acquainted with, if you have not already.

1.1 The natural numbers

We will take the following as our axiomatic definition of the natural numbers. There are constructions of the natural numbers from more basic principles but this is our starting point. As Bertrand Russell (maybe) said: it's turtles all the way down.

Definition 1.2 (The Peano axioms for the natural numbers)**.** The set of *natural numbers*, denoted $\mathbb{N}$, is defined axiomatically by:

1. There is a distinguished element which we denote $1 \in \mathbb{N}$. It is the "first element" of the natural numbers.
2. There is a function $S : \mathbb{N} \to \mathbb{N}$ called the *successor function*. This should be thought of as the function $S(x) = x + 1$. In other words, every element has a (unique) well-defined successor.
3. The first element 1 is not a successor of any element, i.e. 1 is not in the image of S.
4. S is injective, i.e. an element can succeed at most one element.
5. Let P be a property. If 1 has P, and also if x has P implies $S(x)$ has P, then every $x \in \mathbb{N}$ has P. This is the induction principle.

Remark 1.3. The inductive axiom is important. Otherwise, we may be allowed things like two copies of the natural numbers: $\mathbb{N} \cup \mathbb{N}$. In some ways it specifies that there is a unique first element. It does more than that though: it tells us that "arguments by induction are allowed." If your next question is: what other kinds of arguments are allowed or not allowed, my answer is I don't know, and that you should consult a logician. I do not worry much about issues like this.

This is a good starting point, but we are not only interested in $\mathbb{N}$ as a set but also in the addition and multiplication operations on it, which give it some additional structure. We should define these in terms of the axioms above.

Definition 1.4 (Addition, multiplication)**.** One can define addition on $\mathbb{N}$ recursively as follows.

$$a + 1 := S(a)$$
$$a + S(b) := S(a + b)$$

One can also define multiplication on $\mathbb{N}$ rescursively as follows.

$$a \cdot 1 := a$$
$$a \cdot S(b) := a + ab$$

One can then prove that addition and multiplication as defined above are commutative, associative, and distributive, i.e. that the following are true

$$
\begin{aligned}
a + b &= b + a \\
(a + b) + c &= a + (b + c) \\
ab &= ba \\
(ab)c &= a(bc) \\
a(b + c) &= ab + ac
\end{aligned}
$$

Since I am taking the natural numbers as we know them "intuitively" as a starting point, I won't actually prove these things. If you are interested you can try it.

Remark 1.5. Sometimes people define the natural numbers to include zero. I am making an arbitrary decision here not to invite zero to the party. (It wasn't a very positive number to be around... ha ha ha...)

1.2 Equivalence relations

I will introduce a notion which is very useful in making constructions.

Definition 1.6. Given a set S, an *equivalence relation* on the set S is a subset $E \subset S \times S$. If $(a, b) \in E$, we often write $a \sim b$. This subset E satisfies the following properties:
(reflexivity) for all $s \in S$, we have $(s, s) \in E$ (i.e. $s \sim s$),
(symmetry) for all $(s, t) \in E$, we have $(t, s) \in E$ (i.e. if $s \sim t$ then $t \sim s$),
(transitivity) for all $(s, t) \in E$ and $(t, u) \in E$, we have $(s, u) \in E$ (i.e. if $s \sim t$ and $t \sim u$ then $s \sim u$).

This notion is supposed to generalize equality, as equality is very rigid and literal, but there are situations in which we might want to consider two non-equal things to be "the same."

Example 1.7. Examples of equivalence relations:
(1) Equality is an equivalence relation.
e(2) Similarity and congruence in Euclidean geometry.
(3) Similarity of matrices in linear algebra.
(4) Congruence modulo n.

Definition 1.8 (Equivalence classes)**.** We will introduce the following notation first. For $s \in S$, denote $[s]$ to be the subset of S containing all elements equivalent to s, i.e.

$$[s] = \{t \in S \mid s \sim t\}.$$

An *equivalence class* is a subset of the form $[s]$ for *some* $s \in S$.

Remark 1.9 (More intrinsic characterization of equivalence classes)**.** Another characterization of equivalence classes is the following: a subset $T \subset S$ is an equivalence class such that (1) every $t, t' \in T$ has $t \sim t'$; and (2) if $s \notin T$ then for all $t \in T$ we have $t \not\sim s$.

Remark 1.10 (Equivalence classes partition S)**.** Let S be a set with an equivalence relation $\sim$. Then the equivalence classes partition S. Check this as an exercise.

Definition 1.11 (Quotients)**.** Let $\sim$ be an equivalence relation on S. Then the *quotient* $S/\sim$ is the set of equivalence classes.

Example 1.12. Examples of quotients
(1) Under equality, for any set S, $S/\sim = S$.
(2) Left as exercise.
(3) Let M_n be the set of all $n \times n$ matrices. Then $M_n/\sim$ is the set of equivalence classes with representatives all possible Jordan normal forms, up to reordering.
(4) $\mathbb{Z}/\sim = \{[0], [1], \ldots, [n-1]\}$

1.3 The integers

We will define the integers from the natural numbers as follows. The idea is to let a pair (a, b) represent the quantity $a - b$. However, sometimes different pairs will be "equal" under this assignment so we want to capture this using an equivalence relation.

Definition 1.13. The *set of integers* $\mathbb{Z}$ is defined as the quotient $(\mathbb{N} \times \mathbb{N})/\sim$ under the equivalence relation $(a, b) \sim (c, d)$ if $a + d = c + b$. The set of integers has an operation, called *addition*, which is defined by $[(a, b)] + [(c, d)] = [(a + c, b + d)]$. This operation is well-defined in the quotient (check as exercise). The set of integers also has a *additive identity* (i.e. zero), which is $[(1, 1)]$. The set of integers also has an operation called *multiplication* defined by $(a, b) \cdot (c, d) = (ac + bd, ad + bc)$. It also has a *multiplicative identity* given by $[(2, 1)]$. Finally, there is an operation called *additive inversion*, where $-(a, b) = (b, a)$.

Remark 1.14 (Motivating the equivalence relation)**.** We want to think of the pair (a, b) of natural numbers as "representing" the integer $a - b$. However, doing this means that every integer, e.g. 0, has many representatives, e.g. $(1, 1), (2, 2), (3, 3), \ldots$. Thus we must make all these representatives "equal" using an equivalence relation. The equivalence relation that captures when two pairs $(a, b) \sim (c, d)$ is that $a - b = c - d$. However, since subtraction is not well-defined in the natural numbers, we must rewrite this as $a + d = c + b$.

Remark 1.15 (Checking operations are well defined)**.** We defined the addition operation by

$$[(a, b)] + [(c, d)] = [(a + c, b + d)].$$

What we've done is, for each equivalence class, we chose particular representatives and our operation returns back an element of $\mathbb{N} \times \mathbb{N}$. Then we take the equivalence class of that element. To show this is well-defined, one must show that if we chose different representatives, we what we get by applying the operation "rule" gives us an equivalent element.

The set of integers forms what is called a commutative ring.

Definition 1.16. A *commutative ring* is a set R with addition and multiplication operations, denoted $+$ and $\cdot$; additive and multiplicative identities denoted 0 and 1; and additive inverses, denoted $-$. The following properties are satisfied:

1. (associativity of addition) $(a + b) + c = a + (b + c)$
2. (commutativity of addition) $a + b = b + a$
3. (additive identity) $0 + a = a$
4. (additive inverses) $a + (-a) = 0$
5. (associativity of multiplication) $(ab)c = a(bc)$
6. (commutativity of multiplication) $ab = ba$
7. (multiplicative identity) $1 \cdot a = a$
8. (distributivity) $a(b + c) = ab + ac$

Remark 1.17. This is not a very important definition for this course; it appears frequently in algebra. It will not appear beyond this lecture. You can think of this definition to just be a word which is shorthand for saying: a set which has addition and multiplication and their respective identity elements which satisfies commutativity, associativity and distributativity properties. Or, a set that has all the operations that the integers do.

Proposition 1.18. *The integers form a ring.*

Proof. We will prove one of the properties above: additive inverses. The rest are left as exercises. What we want to show is that:

$$(a_+, a_-) + (a_-, a_+) \sim (1, 1)$$

Using the above formula for addition, we have:

$$(a_+ + a_-, a_+ + a_-) \sim (1, 1)$$

This is true, since using our definition of $\sim$, we check that indeed:

$$a_+ + a_- + 1 = a_+ + a_- = 1.$$

□

Example 1.19 (Examples of other commutative rings)**.** The (multivariate) polynomials with integer or rational or real or complex coefficients, under the usual addition and multiplication of polynomials, form a ring, denoted $k[x]$ for one variable, $k[x, y]$ for two variables x and y, et cetera, where $k = \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

1.4 The rational numbers

The construction of the rational numbers is left as a homework exercise. The rational numbers form a field, which is to say all numbers except 0 have a multiplcative inverse.

Definition 1.20. A *field* k is a commutative ring with multiplicative inverses for all nonzero elements, denoted a^{-1}, i.e. such that $aa^{-1} = 1$.

We will make the following definition as well. We could have made similar definitions before, but I am lazy.

Definition 1.21. A (total) *order* on a set S is a subset of $L \subset S \times S$ satisfying the following properties. If $(a, b) \in L$ then we will write for shorthand $a \leqslant b$.
(antisymmetry) if $a \leqslant b$ and $b \leqslant a$ then $a = b$
(transitivity) if $a \leqslant b$ adb $b \leqslant c$ then $a \leqslant c$
(totality) at least one of the following must be true: $a \leqslant b$ or $b \leqslant a$.

An *ordered field* k is a field with a total order such that:
(a) if $a \leqslant b$ then $a + c \leqslant b + c$ for any $c \in k$
(b) if $a \geqslant 0$ and $b \geqslant 0$ then $ab \geqslant 0$

Remark 1.22. There are other notions of orders: for example, there are orders on sets where two elements might be incomparable. You can check Wikipedia if you're interested. We are interested in total orders, but since there are no other orders we will consider, I might be lazy about writing "total."

Example 1.23. The rationals and reals, $\mathbb{Q}, \mathbb{R}$, are both ordered fields. The complex numbers, $\mathbb{C}$ is a field but it is not an ordered field.

2 Lecture 2

2.1 The real numbers by axioms

There are three approaches we will take to the real numbers in this class. The first is axiomatic.

Definition 2.1 (Axiomatic definition of the real numbers)**.** The real numbers are an ordered field with the least upper bound (LUB) property. The LUB property is as follows: every subset which has an upper bound has a supremum (least upper bound). (We will define these terms now).

Definition 2.2. Let X be an ordered set, and $S \subset X$. The *maximum* of S is an element $m \in S$ such that for all $s \in S$, $s \leqslant m$ (exercise: prove maximums must be unique). An *upper bound* on S is an element $x \in X$ such that for all $s \in S$, $s \leqslant x$.

Remark 2.3. Notice that maximums must be in the set S, whereas upper bounds do not have to be. Further, notice that upper bounds are usually not unique.

Example 2.4. Let $X = \mathbb{R}$ and $S = [0, 1]$. Then, 1 is a maximum of S and any number greater than 1 is an upper bound.

Remark 2.5. All finite sets have maximums. However, not all sets in general have a maximum. For a trivial example, consider $(0, \infty) \subset \mathbb{R}$. This fails to have a maximum, and fails to have an upper bound. I can live with this. What's kind of bad is that not all bounded sets have maximums. For example, the interval $(-1, 0)$ is bounded above by 0 but does not have a maximum. Let $m \in (-1, 0)$ be a possible maximum. But, $\frac{1}{2}m$ is also in the set and is even bigger (since m is negative). So there can't be a maximum. In this way asking for maximums is somehow the wrong thing to do for sets which might be infinite. This will motivate the next definition.

Definition 2.6. A *least upper bound* on a subset S of an ordered set X is an element $x \in X$ which is an upper bound on S, and such that there is no smaller upper bound. More mathematically, it is an upper bound such that if x' is another upper bound, the necessarily, $x \leqslant x'$. The least upper bound is also called the *supremum* and is denoted $\sup(S)$. Likewise, the greatest lower bound is called the *infimum* and is denoted $\inf(S)$.

Remark 2.7. In our previous example, the supremum is 0. However, it is still not true that every subset of a well-ordered set has a supremum. For example, consider $(0, \sqrt{2})$ as a subset of $\mathbb{Q}$ (not $\mathbb{R}$!) does not have a supremum. Showing this is actually somewhat involved and involves some elementary number theoretic arguments. I will just appeal to intuition here. In a way, though, this problem is less of a defect of our definition of least upper bound than it is a defect of the rational numbers: the subset $(0, \sqrt{2})$ of the rational numbers doesn't have anything that even resembles a "generalized maximum" if we stay in $\mathbb{Q}$. This motivates the next definition.

Definition 2.8. A well ordered set has the *least upper bound property* if for every bounded above subset S, S has a supremum (least upper bound).

Remark 2.9. The least upper bound property is equivalent to the greatest lower bound property for an ordered field, since for any set S we can always take its negation $-S$, and the supremum of S is the infimum of $-S$ and vice versa.

Remark 2.10. Thus the real numbers are defined as an ordered field which has the LUB property. Vaguely, this means "all its holes are filled in." However, it's still not clear such an object exists, since we haven't constructed it. In the next section we will do that. In a somewhat later section, we will show that there is a general construction one can make to "fill in the holes" on certain kinds of spaces called metric spaces, and that the real numbers are what we get when we do this to $\mathbb{Q}$.

2.2 The real numbers by Dedekind cuts

In this section we will give a construction of the real numbers by Dedekind cuts. The way to "think about" Dedekind cuts is that to a real number α, we associate the two subsets $(-\infty, \alpha)$ and (α, ∞), as subsets of $\mathbb{Q}$ (i.e. intersect with $\mathbb{Q}$). Though this is how we "think" of these cuts, in making arguments we cannot actually refer to things like $(-\infty, \alpha)$ for α irrational, since this would make our construction circular!

Definition 2.11. A *Dedekind cut* of the rational numbers $\mathbb{Q}$ is a partition of $\mathbb{Q}$ into subsets A and B that satisfies the following properties:
(1) (definition of partition) $A \cup B = \mathbb{Q}$ and $A \cap B = \varnothing$
(2) (A is closed downward) if $a \in A$, then $(-\infty, a) \subset A$
(3) (B is closed upward) if $b \in B$¡ then $(b.\infty) \subset B$
(4) (infinity is not a number) neither A nor B are empty nor all of $\mathbb{Q}$
(5) (a number is cut out) A does not have a maximum, and B does not have a minimum

We can give an ordering on the Dedekind cuts as follows. We say $(A, B) \leqslant (A', B')$ if $A \subset A'$ and $B' \subset B$. There is also a way to make Dedekind cuts into a field, but this is left as a homework exercise.

Remark 2.12. Note that the the data of B is extraneous. Given A that satisfies the above properties, it is automatic that $B = \mathbb{Q} - A$ does. If this B has a minimum, then remove it from B. (Exercise: show that this new B cannot have a minimum. Hint: show that between any two rational numbers, there is a rational number.) I made the above definition to illustrate why this is a "cut," but from now on we will only be using the data of A.

Remark 2.13. Property (5) is made because otherwise, the cuts $(-\infty, q)$ and $(-\infty, q]$ for $q \in \mathbb{Q}$ would look like they should be the same. Try to prove the field axioms without this property and see why it's necessary. For example, we can't have two candidates for zero in a field!

Remark 2.14. The Dedekind cuts need to be made into an ordered field. Unfortunately this can get kind of hairy for multiplication, and I won't do it here. Interested readers should refer to the constructions in Rudin.

Proposition 2.15. *The ordered set of Dedekind cuts has the LUB property. In particular, let $\{A_i\}_{i \in I}$ be a collection of Dedekind cuts which is bounded above. Then, its supremum is $\bigcup_{i \in I} A_i$.*

Proof. We need to check that (1) the claimed supremum is also a Dedekind cut (in particular, it is not all of $\mathbb{Q}$) and that (2) it really is the supremum.

For (1), it is easy to see that $\bigcup_i A_i$ is still closed downwards. The fact that the Dedekind cuts are bounded above means there is some cut A such that $A_i \subset A$ for all $i \in I$. This means that $\bigcup_i A_i \subset A$ so $A \neq \mathbb{Q}$. Finally, suppose that $\bigcup_i A_i$ had a maximum, say m. Then $m \in A_p$ for some fixed $p \in I$, but since it is a maximum of the union it would have to be a maximum for A_p as well, contradicting that A_p is a Dedekind cut.

For (2), Suppose there is some Y such that $A_i \subset Y$ for $i \in I$ ($A_i < Y$ for all i, so Y is an upper bound) but $Y \subset \bigcup_{i \in I} A_i$ (Y is lower than the claimed supremum). Let $x \in \bigcup_{i \in I} A_i$; then $x \in A_p$ for some $p \in I$. But this contradicts the first statement, so this cannot be. One can sum this argument up in a sentence like, "the union of a collection of subsets is the minimal subset containing all the subsets in the collection." $\square$

Remark 2.16 (Why we like axioms)**.** This construction is nice in reassuring us that the real numbers exist (yay!). I think the logic-y thing to say would be that Dedekind cuts are a model for the real numbers. The nice thing about axioms is that we can build theories without thinking too hard about specific constructions, as long as our axioms are well-designed. The specific model doesn't really matter as long as we stay within the bounds of the axioms. One might complain that if we want to call something *the* real numbers, there should be a question about uniqueness, which I know nothing about. So this question is also going under the rug. In practice this is not important.

2.3 Properties of $\mathbb{R}$

Proposition 2.17 (Archimedian property)**.** *Let $a, b \in \mathbb{R}$. If $a > 0$ and $b > 0$, then there is a positive integer n such that $na > b$.*

Proof. Suppose otherwise, i.e. that there is some $a, b \in \mathbb{R}$ such that $na \leqslant b$ for all $n \in \mathbb{N}$. In particular, this means that b is an upper bound for $S := \{na \mid n \in \mathbb{Z}\}$. Thus, $\sup S$ is a number.

In a homework exercise, you will show that $\sup(S) + \sup(T) = \sup(S + T)$. Let S be as above and $T = \{a\}$. Then notice that $S + T = S$, and $\sup(T) = a$, so we should have $\sup(S) + a = \sup(S)$ which contradicts that $a > 0$. Thus, the set must be unbounded and there is some n such that $na > b$. $\square$

Proposition 2.18 ($\mathbb{Q}$ is dense in $\mathbb{R}$)**.** *For any $a, b \in \mathbb{R}$, there is a $x \in \mathbb{Q}$ such that $x \in (a, b)$.*

Proof. Write $x = \frac{p}{q}$, for $p, q \in \mathbb{Z}$. Then we want to find p, q such that $aq < p < bq$. We can use the Archimedian property to see that there is an integer q such that $q(b - a) = bq - aq > 1$. We **claim** that if $\beta - \alpha > 1$, then there is an integer in (α, β). This would complete the proof, letting $\beta = bq$ and $\alpha = aq$, and p the integer. $\square$

Lemma 2.19. *Let $\alpha, \beta \in \mathbb{R}$. If $\beta - \alpha > 1$, then there is an integer $p \in (\alpha, \beta)$.*

Proof. If $\alpha < 0$ and $\beta > 0$, we can prove this result for the smaller interval $(0, \beta)$. If $\alpha, \beta < 0$, we can prove this result for the interval $(-\beta, -\alpha)$. So, without loss of generality, we can assume that $\alpha, \beta \geqslant 0$:

The strategy is as follows. We want to take the set

$$\{n \in \mathbb{Z} \mid \alpha < n\}$$

and find its minimum value, say p. By construction we have $\alpha < p$ and since it is the minimum, we kno wthat $p - 1 < \alpha$ and so $p < \alpha + 1 < \beta$ as well, completing the proposition. The issue is that this set may not have a minimum and it may also be empty.

First, let's address the existence of a minimum. We can force a set to have a minimum by making it finite, i.e. by adding some conditions bounding the set:

$$\{n \in \mathbb{Z} \mid \alpha < n \text{ and } n \in [0, N]\}$$

for some integer N. This is true so long as $\alpha < N$, and we can use the Archimedian property to find an integer N such that $N \cdot 1 > \max(|\alpha|)$. □

3 Lecture 3

3.1 Metric spaces

Definition 3.1. Let k be a totally ordered field. A *k-metric space* is a set X along with a *distance function* $d : X \times X \to k$ satisfying the following properties:
(non-negativity) $d(x, y) \geqslant 0$ for all $x, y \in X$,
(identity of indiscernables) $d(x, y) = 0$ if and only if $x = y$,
(symmetry) $d(x, y) = d(y, x)$ for all $x, y \in X$,
(triangle inequality) $d(x, y) + d(y, z) \geqslant d(x, z)$ for all $x, y, z \in X$.

Remark 3.2. Most texts fix $k = \mathbb{R}$. In general, we will assume $k = \mathbb{R}$, but later when we give a consruction of $\mathbb{R}$ we will require $k = \mathbb{Q}$ to avoid a circular argument.

Remark 3.3. I should be more careful about this definition than I am. I am not sure whether any completeness properties of $\mathbb{R}$ are used in results in the literature on metric spaces. I am also not sure how much in the literature applies to loosening the restriction on k, i.e. if we consider ordered abelian groups or ordered rings instead. I hope that these notes at least are internally consistent.

Example 3.4 (The Euclidean metric)**.** $\mathbb{R}^n$ is a metric space under the Euclidean metric. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$, and define:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

One can think of this is a "generalized Pythagorean theorem" way of finding distance. For $n = 1$ it is not hard to see this is a metric. For larger n, the proofs I know use Cauchy-Schwartz. Note that this metric doesn't quite make sense when $k = \mathbb{Q}$ for $\mathbb{Q}^n$ when $n \neq 1$, since square roots might not exist in $\mathbb{Q}$.

Example 3.5 (The discrete metric)**.** Let X be any set. Then the discrete metric is defined by:

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

One can easily verify this is a metric.

3.2 Topological definitions

Definition 3.6. Let (X, d) be a metric space. The *open ball of radius ϵ at x* is defined by:

$$B_\epsilon(x) := \{p \in X \mid d(x, p) < \epsilon\}$$

and the *closed ball* is defined by:

$$\overline{B}_\epsilon(x) := \{p \in X \mid d(x, p) \leqslant \epsilon\}.$$

We will sometimes use the word *neighborhood* of x to refer to any open ball of any radius at x.

Example 3.7. Let $X = \mathbb{R}$ with the Euclidean metric. Then $B_\epsilon(x) = (x - \epsilon, x + \epsilon)$.

Example 3.8. Let X have the discrete metric. Then $B_\epsilon(x) = \{x\}$ if $\epsilon < 1$ and $B_\epsilon(x) = X$ otherwise.

Definition 3.9. Let (X, d) be a metric space, and $E \subset X$. A point $p \in E$ is called an *interior point* or E if there is a neighborhood U of x such that $U \subset E$. A subset E is called *open* if every point is interior.

Example 3.10 (Open balls are open)**.** Every open ball is open. To show that every $p \in B_\epsilon(x)$ is an interior point, take $a = \epsilon - d(p, x) > 0$ note that by the triangle inequality, $B_a(p) \subset B_\epsilon(x)$.

Definition 3.11. Let (X, d) be a metric space and $E \subset X$. A point $p \in X$ is a *limit point* of E if every neighborhood U of p has a point $q \neq p$ such that $q \in E$. A subset E which contains all its limit points is called *closed*. The *closure* of E, denote $\overline{E}$, is the union of E with all of its limit points.

Example 3.12 (Closed balls are closed)**.** Every closed ball is closed. Consider $\overline{B}_\epsilon(x)$ and let p be a possible limit point. If $d(p, x) \leqslant r$ then we are happy. Otherwise, let $a < d(p, x) - r$. Then $B_a(p)$ and $\overline{B}_\epsilon(x)$ are disjoint by the triangle inequality, and thus p cannot be a limit point.

Proposition 3.13 (The closure of any subset is closed)**.** *Let (X, d) be a metric space and $E \subset X$. The closure $\overline{E}$ is closed.*

Proof. This might seem obvious, but there is something to prove. One has to show that one does not introduce more limit points by including the limit points of E. Suppose p is a limit point of $\overline{E}$; then every neighborhood N of p intersects $\overline{E}$. If N contains a point of E we are done. If N contains a limit point of E, say q, then since N is open, q is an interior point, so there is a neighborhood of q contained in N which contains a point of E, concluding the proof. □

Remark 3.14. The closed ball isn't really used that much as a topological notion, as far as I know. I include it here for fun.

Definition 3.15. Let (X, d) be a metric space. A subset $E \subset X$ is *dense* if $\overline{E} = X$.

Example 3.16. As we've shown before, $\mathbb{Q}$ is dense in $\mathbb{R}$. To see why, we claim that every real number is a limit point of rational numbers: let $a \in \mathbb{R}$; every ball $B_\epsilon(a) = (a - \epsilon, a + \epsilon)$ contains a rational number.

3.3 Some topological fundamentals

Proposition 3.17 (Complements)**.** *A set is open if and only if its complement is closed.*

Proof. This proof is mostly just juggling the logic in the definitions. Suppose that E is open. Let p be a limit point of E^c. Then every neighborhood of p intersects E^c nontrivially, i.e. every neighorhood is not contained in E, so $p \notin E$ since it cannot be an interior point, so $p \in E^c$, so E^c is closed. Suppose that E is closed. Let $p \in E^c$, so p is not a limit point of E, so their is a neighborhood of p disjoint from E, so p is an interior point, so E^c is open. □

Proposition 3.18 (Unions and intersections)**.** *Any (possibly infinite) union of open sets is open. Any (possible infinite) intersection of closed sets is closed. Any finite union or intersection of open or closed sets is open or closed, respectively.*

Proof. By the previous proposition, we only need to prove the statements for open sets. I will leave the statement that an infinite union of open sets is open as an exercise. For a finite intersection of open sets, take open sets $G_1, \ldots, G_n$, and $x \in \bigcap_{i=1}^n G_i$. Since $x \in G_i$ for all i is interior, we have $B_{\epsilon_i}(x) \subset G_i$ for all i and for some ϵ_i. Then, $B_{\min(\epsilon_i)}(x) \subset \bigcap_{i=1}^n G_i$ and the result follows. □

Proposition 3.19 (Closure)**.** *The closure $\overline{E}$ is the intersection of all closed sets that contain E. Equivalently, it is the smallest closed set containing E.*

Proof. Let $F = \bigcap_{E \subset V closed} V$ be the intersection of all closed sets containing E. Clearly, $F \subset \overline{E}$ since $\overline{E}$ is closed and contains E. We want to show that every closed set containing E also contains $\overline{E}$. Let V be closed and $E \subset V$. Taking the closure of both sides, we find that $\overline{E} \subset V$. □

4 Lecture 4

4.1 Sequences and convergence

Remark 4.1. Warning: the sequence notion of limit is not quite the same as than the notion of limit you learned in calculus – the context is different.

Definition 4.2. Let (X, d) be a metric space. A sequence $\{p_n\}$ in X *converges* to $p \in X$ if for every $\epsilon > 0$, there is some N such that $d(p_n, p) < \epsilon$ when $n > N$. Equivalently, $\{p_n\}_{n \geqslant N} \subset B_\epsilon(x) \cap \{p_n\}$.

In this case, we write:

$$\lim_{n \to \infty} p_n = p$$

If a sequence does not converge to any point, it diverges.

Proposition 4.3. *Limits of sequences are unique.*

Proof. Let p, q be two limits of a sequence $\{p_n\}$. We will show that $d(p, q) < \epsilon$ for any $\epsilon > 0$. Choose any $\epsilon > 0$; then there is an N such that $d(p_n, p) < \frac{\epsilon}{2}$ for $n > N$ and an M such that $d(q_n, p) < \frac{\epsilon}{2}$ for $n > M$. Then, for $n > \max(N, M)$, we have that $d(p, q) \leqslant d(p, p_n) + d(p_n, q) < \epsilon$ as desired. □

Proposition 4.4 (Connections to topology)**.** *If $E \subset X$ and p is a limit point of E, then there is a sequence in E converging to p.*

Proof. For every n, choose some $s_n \in B_{1/n}(p) \cap E$, which we know is nonempty since p is a limit point of E. We claim this sequence converges to p. For $\epsilon > 0$, choose N such that $\frac{1}{N} < \epsilon$. Then, one sees that for $n > N$, we have $d(s_n, p) < \frac{1}{N} < \epsilon$. □

Definition 4.5. Let (X, d) be a metric space. A subset $E \subset X$ is *bounded* if there is a number L such that $\{d(p, q) \mid p, q \in E\}$ is bounded above (in the order-theoretic sense) by L.

Proposition 4.6 (Convergent sequences are bounded)**.** *Convergent sequences are bounded (considered as sets).*

Proof. Let $\{p_n\} \to p$ be the convergent sequence. We will break up the sequence into two "groups" for which we can find a bound on $d(p_n, p_m)$: one for $m, n \leqslant N$ and one for $m, n \geqslant N$. Fix any ϵ (for example, $\epsilon = 1$) and take an N such that $n \geqslant N$ implies $d(p_n, p) < \epsilon$. By the triangle inequality, $d(p_n, p_m) < 2\epsilon$ for $n, m \geqslant N$. This is the first "group." For the second, we know that for $n, m \leqslant N$, $d(p_n, p_m) \geqslant m := \max(\{d(p_n, p_m\} \mid n, m \leqslant N\}$. Then, using the triangle inequality, we know that

$$d(p_n, p_m) \leqslant d(p_n, p_N) + d(p_N, p_m) \leqslant 2\max(m, 2\epsilon)$$

for *any* m, n, establishing a bound. □

4.2 Sequences in $\mathbb{R}$

Proposition 4.7. *Let $\{s_n\}$ and $\{t_n\}$ be sequences in $\mathbb{R}$, with limits s and t respectively. Let $c \in \mathbb{R}$ be a constant. Then:*
(a) $\lim_{n\to\infty}(cs_n) = cs$ *and* $\lim_{n\to\infty}(c + s_n) = c + s$.
(b) $\lim_{n\to\infty}(s_n + t_n) = s + t$
(c) $\lim_{n\to\infty}(s_n t_n) = st$
(d) $\lim_{n\to\infty}(s_n/t_n) = s/t$ *if for all n, $s_n \neq 0$ and $s \neq 0$*

Proof. The proof of (a) is easy and left as an exercise. For (b), fix an $\epsilon > 0$. We want to find an N such that if $n > N$ then $|(s + t) - (s_n + t_n)| < \epsilon$. By the triangle inequality, we have that

$$|(s + t) - (s_n + t_n)| \leqslant |s - s_n| + |t - t_n|.$$

By the limits we already know, we can find an N_s such that $n > N_s$ implies $|s - s_n| < \frac{1}{2}\epsilon$ and likewise an N_t. Then, let $N = \max(N_s, N_t)$ and we have the result.

For (c), rewrite the expression:

$$st - s_n t_n = (t - t_n)(s + s_n) + t_n s - s_n t = (t - t_n)(s + s_n) + (t_n - t)s - (s_n - s)t$$

The strategy is as follows: in each of the three additive terms, we have one factor that should "go to zero." We also have bounds on the other factors, which guarantees that their products "go to zero." More precisely let L be a bound on the sequence $\{s + s_n\}$. Fix $\epsilon > 0$; we want to find an N such that if $n > N$ then $|st - s_n t_n| < \epsilon$. Applying the triangle inequality to our rewriting, we find that

$$|st - s_n t_n| \leqslant |L||t - t_n| + |s||t - t_n| + |t||s - s_n|$$

There is an N such that for $n > N$, we have $|s - s_n| < \frac{\epsilon}{|L|+|s|+|t|}$ and $|t - t_n| < \frac{\epsilon}{|}L| + |s| + |t|$. Then, one finds that $|st - s_n t_n| < \epsilon$ as desired.

For (d), we only need to show that

$$\lim_{n\to\infty} \frac{1}{s_n} = \frac{1}{s}.$$

Fix $\epsilon > 0$. We want to find N such that for $n > N$, we have:

$$|\frac{1}{s_n} - \frac{1}{s}| = |\frac{s_n - s}{s_n s}| < \epsilon$$

The idea is as follows: the denominator $s_n s$ tends toward s^2 which is a constant we can account for. We want to "overestimate" this quantity with respect to s^2 since it is in the denominator. So, we can find N_1 such that $n > N_1$ implies that $2|s_n| > |s|$ (check this), and so $|s_n s| > \frac{1}{2}s^2$. Thus our expression becomes:

$$|\frac{s_n - s}{s_n s}| < 2\frac{1}{s^2}|s_n - s|$$

Now, we can find N_2 such that $|s_n - s| < \frac{s^2\epsilon}{2}$ and the result follows. □

4.3 Extended real numbers

Definition 4.8. We write

$$\lim_{n\to\infty} s_n = \infty$$

to mean that for each M there is some N such that $n > N$ implies that $s_n > M$. Likewise, we write

$$\lim_{n\to\infty} s_n = -\infty$$

to mean that for each M there is some N such that $n > N$ implies that $s_n < N$.

Remark 4.9. The "extended reals" are not a metric space, nor a field, though it does have an total order. However, the following arithmetical sentences still "make sense" for any $x \in \mathbb{R}$:
$x + \infty = \infty$, $x - \infty = -\infty$, $\frac{x}{\infty} = \frac{x}{-\infty} = 0$
if $x > 0$ then $x \cdot \infty = \infty$ and $x \cdot (-\infty) = -\infty$
if $x < 0$ then $x \cdot (-\infty) = \infty$ and $x \cdot \infty = -\infty$
The statements of the above proposition are still true in these cases.

5 Lecture 5

5.1 Compactness

Remark 5.1 (Induced metric)**.** Let (X, d) be a metric space, and let $Y \subset X$. Then, $(Y, d|_Y)$ is a metric space, where $d|_Y$ is the distance function when restricted to Y.

Proposition 5.2. *Let (X, d) be a metric space, and $E \subset X$. Let U be open in X. Then $U \cap E$ is open in E.*

Proof. This is left as an exercise. Just check the definition, and note that open balls in Y are open balls in X intersected with Y. □

Remark 5.3 (Openness and closedness as properties of subsets)**.** Asking whether a set is opened or closed only makes sense once we identify it as an *subset* of a metric space, and the answer depends on its embedding as a subset. For example, $[0, 1]$ is an open subset of the metric space $[0, 1]$, but it is not an open subset of the metric space $\mathbb{R}$. The notion we will now introduce, compactness, is intrinsic: it does not depend on any embedding.

Definition 5.4. Let (X, d) be a metric space. Then, X is *compact* if every open cover has a finite subcover. An *open cover* of a space X is a (possibly infinite) collection of open subsets of X, say $\{U_i\}$ such that they cover X, i.e. $\bigcup U_i = X$. A *finite subcover* is a finite subcollection which is still a cover.

Example 5.5 (Open covers of $\mathbb{R}$)**.** Here are many many examples of open covers of $\mathbb{R}$. Here are some.
(1) $U_n = (-n, n)$ for $n \in \mathbb{Z}$,
(2) $U_n = (-\infty, n)$ for $n \in \mathbb{N}$,
(3) $U_n = (-\infty, \infty)$ for $n \in \mathbb{Z}$,
(4) $U_n = (a + 1, a - 1)$ for $a \in \mathbb{R}$,
(5) $U_n = (a + \epsilon, a - \epsilon)$ for $a, \epsilon \in \mathbb{R}$.
I claim that none of these have finite subcovers. I'll prove this for (1) and leave the rest as an exercise.

Proof. Suppose there was a finite subcover; then it would be given by a finite subset of the index set of integers, call it $I \subset \mathbb{Z}$. Check that $\bigcup_{i \in I}(-n, n) = (-|\max(I)|, |\max(I)|)$, which is not $\mathbb{R}$. □

Example 5.6 (Not open covers)**.** Let $X = (0, 1] \subset \mathbb{R}$. Then $(0, \frac{1}{n})$ is not an open cover, since $1 \in X$ is not in any of those open sets.

Corollary 5.7. *$\mathbb{R}$ is not compact.*

Proof. To be compact, *every* open cover must have a finite subcover. Since we've given an open cover which does not have a finite subcover, we are done. □

Remark 5.8 ($\mathbb{R}$ has an open cover with a finite subcover)**.** Note that example (3) is an open cover which has a finite subcover. This does not contradict that $\mathbb{R}$ is not compact, since only *one* open cover can break compactness.

Example 5.9 (Compact intervals in $\mathbb{R}$)**.** Some more examples which we can do explicitly.
(1) $(-1, 1)$ is not compact.
(2) $[0, 1]$ is compact.
(3) $(0, \infty)$ is not compact.
(4) $[0, \infty)$ is not compact.
(5) $\mathbb{Z}$ is not compact.
(6) $\mathbb{Q} \cap [0, 1]$ is not compact.

Proof. (1) Use the open subcover given by $U_n = (-1 + \frac{1}{n}, 1 - \frac{1}{n})$ for $n \in \mathbb{N}$.
(2) We will prove a general theorem later.
(3) Use the open subcover $(0, n)$ for $n \in \mathbb{N}$.
(4) Use the open subcover $[0, n)$ for $n \in \mathbb{N}$. Note that while these are not open in $\mathbb{R}$, they are open in $[0, \infty)$.
(5) Use the open subcover $\{n\}$ for $n \in \mathbb{Z}$.
(6) This is a little trickier. Choose an irrational number $\alpha \in [0, 1]$. Let s_n be a rational number in $(\alpha - \frac{1}{n}, \alpha)$. Then, take the open subcover $\mathbb{Q} \cap ([0, s_n) \cup (\alpha, 1])$ for $n \in \mathbb{N}$. □

Proposition 5.10. *Let (X, d) be a metric space and $E \subset X$. If E is compact then E is a closed and bounded subset of X.*

Proof. We can do the following for any metric space. For every $x \in E$ (and $x \neq p$), take a neighborhood U_x which is disjoint from some neighborhood of p. For example, one could take $U_x = B_{\frac{d(x,p)}{2}}(x) \cap E$. If $p \neq E$, this is an open cover, since it contains every point in E. By compactness, take a finite subcover, say indexed by the finite points $x_1, \ldots, x_n \in E$. Then, notice that $\bigcup_{i=1}^n U_{x_n}$ is disjoint from $B_{\min(\frac{d(x_i,p)}{2})}(p)$, so p is not a limit point. Thus E contains all of its limit points.

For boundedness, let $x \in E$ be any point, and let consider the open cover $U_n = B_n(x) \cap E$. By compactness this has an open subcover, and let N be the maximum n in this subcover. Then $E \subset B_N(x)$ and is bounded. □

Remark 5.11. It is a little remarkable that an intrinsic property of a space tells us a property about *every* embedding of that space.

Proposition 5.12. *Let X be a compact metric space. A closed subset of X is compact.*

Proof. Let $E \subset X$ be closed. Take a finite subcover U_i of E. First, we claim (and will prove later) that an open set of E can be written as the intersection of E with an open set of X. Given this claim, for each U_i take $V_i \subset X$ open such that $U_i = V_i \cap E$. Now take the open cover $\{V_i\} \cup \{X - E\}$ of X. Since X is compact, this has a finite subcover – if $X - E$ is in the finite subcover, remove it to obtain a finite subcover of E. □

Lemma 5.13. *Let (X, d) be a metric space and let $E \subset X$ be a subset considered as a metric space. Let U be an open set of E. Then there is an open set V of X such that $U = E \cap V$.*

Proof. Recall that every open set can be written as the union of open balls. Take the same open balls but in X instead. □

Proposition 5.14. *Any finite union of compact subsets is compact.*

Proof. Left as an exercise. □

6 Lecture 6

6.1 Compactness in $\mathbb{R}^k$

We will make the following definition, which will only be used in this section.

Definition 6.1. A *k-cell* is a subset of $\mathbb{R}^k$ of the form

$$[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_k, b_k]$$

i.e. a "box" with given vertices.

Lemma 6.2 (Cantor intersection lemma for $\mathbb{R}^k$)**.** *Let $E_0 \supset E_1 \supset E_2 \supset \cdots$ be a descending chain of nested nonempty k-cells of $\mathbb{R}^k$. Then, $\bigcap_i E_i$ is nonempty.*

Proof. First, we can prove this for $\mathbb{R}^1$. Consider a sequence where $E_i = [a_i, b_i]$. To be nested means $a_1 \leqslant a_2 \leqslant \cdots$ and $b_1 \geqslant b_2 \geqslant \cdots$. Let $a = \sup(a_1, a_2, \ldots)$ and $b = \inf(b_1, b_2, \cdots)$. We claim that $a \leqslant b$. Then any $x \in [a, b]$ would suffice. To prove the claim, suppose that $a > b$; let $x \in (b, a)$. Then there is some n for with $x < a_n$ and also some m for with $x > b_m$. Then, $b_{\max(m,n)} < x < a_{\max(m,n)}$, which cannot happen. To prove this for $\mathbb{R}^k$, one makes a similar argument for each component of the k-tuple, and is left as an exercise. □

Theorem 6.3 (Heine-Borel theorem)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. We've already proven one direction. Now let $X \subset \mathbb{R}^n$ be closed and bounded. Since X is bounded, it is contained in a closed box, $T_0 := [-N, N]^n$ for large N. Since a closed subset of a compact set is compact, it suffices to show that this box is compact.

Fix an open cover of the box T_0. If one cuts this box in half in all possible directions, one obtains 2^n smaller boxes with half the side length of the original. If every one of the 2^n half-boxes had a finite subcover, then T_0 would have a finite subcover (by taking the finite union of the finite subcovers). So, suppose otherwise, and choose a box T_1 which does not have a finite subcover. We can repeat this process to obtain a descending chain of nested boxes:

$$T_0 \supset T_1 \supset T_2 \supset \cdots .$$

By the previous lemma, this is nonempty and contains a point p. This point p must be in one of the open sets of the cover, say U, and thus there is some ball $B_\epsilon(p) \subset U$. But, this ball contains all T_i for $i > M$ for some large M, contradicting that each of the T_i does not have a finite subcover. This proves the theorem. □

Example 6.4 (Not true for all metric spaces)**.** The Heine-Borel theorem is not true for all metric spaces. For example, if X is an infinite metric space with the discrete topology, then a subset is compact if and only if it is finite. However, every subset is closed and bounded.

Remark 6.5. This theorem might convince you that compactness is a redundant notion. In the following sections we will see that this is very untrue, if we want to work in the general context of metric spaces or topological spaces.

7 Lecture 7

7.1 Subsequences

I will not give a definition of subsequences, and convergence of a subsequence. It is exactly what you think it is. However, we will make a few definitions anyway.

Definition 7.1. A sequence s in $\mathbb{R}$ is *monotonically increasing* if $s_i \leqslant s_{i+1}$ for all i. It is *monotonically decreasing*

Proposition 7.2. *All bounded monotone sequences converge.*

Proof. Suppose s_n is monotonically increasing. We claim s_n converges to $s := \sup(\{s_n\})$. For $\epsilon > 0$, there is some $s_N \in (s - \epsilon, s)$ by definition of supremum. Since the sequence is mootonic, this is true for all s_n and $n > N$. □

Proposition 7.3. *Every sequence has a monotone subsequence.*

Proof. Let s_n be a sequence. We say the kth term, s_k, is *dominant* if $s_k > s_i$ for all $i > k$, i.e. bigger than all its subsequent terms. There are two cases to consider. In the first case, there are infinitely many dominant terms; then, the subsequence consisting of dominant terms is a monotonically decreasing sequence. In the second case, there are only finitely many. Then, in choosing a subsequence, we follow the rule: never choose a dominant term. In doing so, we can always choose some term in the sequence following any choice which is bigger, obtaining a monotonically increasing sequence. □

Theorem 7.4 (Bolzano-Weierstrass for $\mathbb{R}$)**.** *Every bounded sequence has a convergent subsequence.*

Proposition 7.5 (Bolzano-Weierstrass, generalization)**.** *Let X be a compact metric space. Every sequence has a convergent subsequence.*

Proof. Let $\{x_n\}$ be the sequence, and let $E = \{x_1, x_2, \ldots\}$ i.e. the set of points in the sequence. If E is finite then there must be a point that reoccurs infinitely many times, and that is our convergent subsequence. If E is infinite, then we claim it has a limit point, say p. Given this claim, for each ball $B_{\frac{1}{n}}(p)$ we can choose successive elements in smaller balls converging to p. More precisely, there is some N_n such that s_i is in the ball $B_{\frac{1}{n}}(p)$ for $i > N_n$. Choose successively i_n such that $i_n > i_{n-1}$ and $i_n > N_n$ to obtain the subsequence. □

Lemma 7.6. *If E is an infinite subset of a compact metric space X, then E has a limit point in X.*

Proof. Suppose to the contrary; if no point in X is a limit point, every point in $x \in X$ has a neighborhood which contains at most one point of E (namely, the point x itself if $x \in E$). This gives an open cover of X, but it cannot have a finite subcover since each open contains at most one point of E. □

Remark 7.7. Note that this proposition is not true for just closed and bounded subsets. For example, take $X = \mathbb{Z}$ under the discrete topology, so that it is closed and bounded but not compact. The sequence $s_n = n$ has no convergent subsequence. So this is another piece of evidence in favor of compactness as a "good notion."

Definition 7.8. The *limit supremum* of a sequence s_n is defined as

$$\limsup_{n\to\infty} = \lim_{n\to\infty} \sup(s_n, s_{n+1}, s_{n+2}, \ldots)$$

The *limit infimum* is defined similarly

$$\liminf_{n\to\infty} = \lim_{n\to\infty} \inf(s_n, s_{n+1}, s_{n+2}, \ldots)$$

Proposition 7.9 (Characterization of limit supremum)**.** *Let s_n be a bounded sequence. Let E be the set of subsequential limits. Then* $\limsup s_n$ *exists and is equal to* $\sup(E)$*, and* $\liminf s_n$ *exists and is equal to* $\inf(E)$*.*

Proof. I will prove the statement for lim sup. Since s_n is bounded above, $\sup(E)$ exists. **ehhhhhhhhh** □

Proposition 7.10. *For any sequence s_n we have that* $\liminf s_n \leqslant \limsup s_n$*. If they are equal, then the limit exists and* $\lim_{n\to\infty} s_n = \liminf s_n = \limsup s_n$*. Conversely, if the limit exists, then* $\lim_{n\to\infty} s_n = \liminf s_n = \limsup s_n$*.*

Proof. The first statement is left as an exercise. For the second, let $s = \limsup s_n = \liminf s_n$. Fix $\epsilon > 0$. There is an N such that for $n > N$, we have that $\sup(s_n, s_{n+1}, \ldots) \in (s - \epsilon, s + \epsilon)$. Thus, $s_n < s + \epsilon$ for all $n > N$. A similar argument with the infimum shows that $s - \epsilon < s_n$ for all $n > N$. Thus, $|s_n - s| < \epsilon$ for $n > N$. For the third statement, we have that for every $\epsilon > 0$, there is some N such that for $n > N$ we have $s_n \in (s - \epsilon, s + \epsilon)$. Thus, $\limsup(s_n) < s + \epsilon$ and $\liminf > s - \epsilon$ for every ϵ, and the result follows. □

Corollary 7.11 (Application to sequences: "squeezing")**.** *Let s_n, t_n be sequences in $\mathbb{R}$ that converge to s, t, such that $s_n \leqslant t_n$ for $n > N$ for some N. Then,* $\lim s_n \leqslant \lim t_n$*.*

Proof. One can easily check that $\limsup s_n \leqslant \liminf t_n$. The rest follows by the previous proposition. □

7.2 Cauchy sequences, complete metric spaces

Example 7.12. Let $X = \mathbb{R} - \{0\}$ be a metric space. The sequence $\{\frac{1}{n}\}$ does not converge, because $0 \notin X$. But it "looks like" a convergent sequence. We want to make a definition that captures this phenomenon.

Definition 7.13. A sequence is *Cauchy* if for every $\epsilon > 0$ there is an integer N such that $d(p_n, p_m) < \epsilon$ for $m, n > N$. A metric space for which every Cauchy sequence is convergent is called *complete*.

Remark 7.14. As we saw above, not every Cauchy sequence converges, and in our example was because there was a "hole."

Proposition 7.15. *Cauchy sequences are bounded*

Proof. Left as an exercise. □

Proposition 7.16. *Every convergent sequence is Cauchy.*

Proof. Suppose s_n converges to s. Then, for every $\epsilon > 0$ there is an N such that $d(s_n, s) < \frac{\epsilon}{2}$ for $n > N$. Then, for $m, n > N$, $d(s_n, s_m) \leqslant d(s_n, s) + d(s_m, s) < \epsilon$. □

Proposition 7.17 (Cantor intersection lemma)**.** *Let $E_0 \supset E_1 \supset E_2 \supset \cdots$ be a descending chain of nonempty compact subsets of a metric space X. Then, $\bigcap_i E_i$ is nonempty. Further, define the* diameter *of a set E by*

$$\operatorname{diam}(E) := \sup(\{d(x, y) \mid x, y \in E\}.$$

If $\lim_{n\to\infty} \operatorname{diam}(E_i) = 0$, then $\bigcap E_i$ consists of exactly one point.

Proof. For the first statement, if the intersection is empty, then $\{E_0 - E_i\}_{i\in\mathbb{N}}$ is an open cover of E_0 (open because E_i is compact in compact E_0 and thus closed). It has a finite subcover, since E_0 is compact. This means that eventually the sequence stabilizes, i.e. $E_i = E_{i+1}$. This contradicts that the E_i are nonempty, so the intersection must be nonempty. For the second statement, this must be true since if the intersection contained two points, say x, y, then $\operatorname{diam}(E_i) \geqslant d(x, y)$ for all i. □

Proposition 7.18. *Every compact space is complete.*

Proof. Let s_i be a Cauchy sequence. Define $E_i = \overline{\{s_i, s_{i+1}, \ldots\}}$, in the notation of the previous proposition. Since the sequence is Cauchy, $\lim_{n\to\infty} \operatorname{diam}(E_n) = 0$. Thus there is a single point in the intersection, say s. I claim this is the limit (the "hard part" is over, since the failure for a Cauchy sequence to converge is somehow due to the point it should converge to "not being there"). To see this, fix $\epsilon > 0$; if $\operatorname{diam}(E_N) < \epsilon$ then this implies that $d(s_n, s) < \epsilon$ for all $n > N$. □

Corollary 7.19. *$\mathbb{R}^k$ is complete.*

Proof. Consider a Cauchy sequence; it is bounded (check as exercise) so we may enclose it in a bounded and closed box. This is compact by Heine-Borel, and thus every bounded Cauchy sequence has a limit. □

Definition 7.20. Let (X, d) be a metric space. We will define the *completion* of X, denoted $\overline{X}$. As a set, the elements of $\overline{X}$ are Cauchy sequences (indexed by $\mathbb{N}$) in X, modulo the an equivalence relation which we will define later. For now, define the distance function between two Cauchy sequences by:

$$d(x, y) = \lim_n d(x_n, y_n)$$

Check as an exercise that this is well-defined since the sequences are Cauchy. Then, two sequences are equivalent $x \sim y$ if $d(x, y) = 0$. This distance function is also well-defined on equivalence classes and satisfies the axioms of a distance function.

7.3 Aside: Construction of the real numbers by completion

In this section we will backtrack and pretend we don't know what $\mathbb{R}$ is. We will treat $\mathbb{Q}$ as a $\mathbb{Q}$-metric space. This will be the only time in these notes that we think of distances as taking values in any field other than $\mathbb{R}$. We will use Cauchy sequences in $\mathbb{Q}$ to construct $\mathbb{R}$.

Definition 7.21. The set of *real numbers* is defined to be the set of Cauchy sequences in $\mathbb{Q}$ (using the absolute value metric) modulo the equivalence relation where $x \sim y$ if $\lim_{n\to\infty} x_n - y_n = 0$. Addition and multiplication are defined on Cauchy sequences by $(x + y)_n = x_n + y_n$ and likewise. The additive identity is the constant zero sequence, and the multiplicative identity is the constant one sequence. The order is defined by $x \leqslant y$ if $x \sim y$ of if there is some N such that $x_n < y_n$ for all $n > N$. Finally, the rational numbers are considered as a subset of $\mathbb{R}$ by taking constant sequences.

Proof. Check that all operations and orders are well-defined and the field axioms as an exercise. One is assigned as homework.

The upper bound property is a little more work. Let S be a subset of real numbers, and let u_0 be an upper bound of S. Choose ℓ_0 such that there is some $s \in (\ell_0, u_0) \cap S$. One should quickly check that this can be done: take some $s \in S$ and find the N such that $m, n > N$ implies $|s_n - s_m| < 1$, and we can take $\ell_0 = s_n - 2$. Now, repeat the following process to construct a sequence: consider $m_n = \frac{\ell_{n-1}+u_0}{2}$. If this is an upper bound for S, then set $u_n = m_n$ and $\ell_n = \ell_{n-1}$. Otherwise, set $u_n = u_{n-1}$ and $\ell_n = m_n$. This defines two sequences, u_n and ℓ_n. These are both Cauchy sequences; I will leave this as an exercise.

First, notice that every u_n is an upper bound on S by construction. Thus the sequence u *thought of as an element* of $\mathbb{R}$ is an upper bound on S by definition. I claim that u is the supremum. To see this, suppose there is a smaller upper bound, u'. This means that $\lim_{n\to\infty} u_n - u'_n = a$ for some positive number a. Notice that the the quantity $u_n - \ell_n$ is cut in half each time we increase n by one. Also notice that by construction, there is always a element in S greater than ℓ_n. Thus, when $u_n - \ell_n < a$, we find that this contradicts that u'_n is an upper bound. □

8 Lecture 8

8.1 Taking powers in the real numbers

We haven't actually define what it means to take powers in the *positive* real numbers to numbers other than integers. We will first define fractional powers, which we could have done earlier as soon as we defined Dedekind cuts.

Definition 8.1. Let $\alpha \in \mathbb{R}$ be have $\alpha > 1$ and $p \in \mathbb{N}$. We define $\alpha^{1/p}$ to be the unique positive real number x such that $x^p = \alpha$. Constructively, this is the Dedekind cut:

$$\{x \in \mathbb{Q} \mid x \leqslant 1 \text{ or } x^p < \alpha\}.$$

Now, let $p \in \mathbb{R}$ be positive, and $\alpha > 1$. We define α^p to be the Dedekind cut:

$$\{x \in \mathbb{Q} \mid x \leqslant 1 \text{ or } x^q < \alpha \text{ for } 0 < q < p\}.$$

For the case $0 < \alpha < 1$, one uses that $\alpha^{-1} > 1$. One does have to prove that this cut does have the given property, and that it is unique in this sense. The first is left as an exercise, and the second follows from the order axioms.

We won't prove the following.

Proposition 8.2. *For $a, b > 0$ and p real numbers, using the above definitions, we have that $(ab)^p = a^p b^p$.*

8.2 "Toolbox" sequences

Recall:

Theorem 8.3 (Binomial theorem)**.** *Let $n \in \mathbb{N}$. Then,*

$$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + nx^{n-1} + x^n$$

Proposition 8.4. *The following are useful sequences to know.*
(a) If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.
(b) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
(c) $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
(d) If $p > 0$, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
(e) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

Proof. For (a), we want to verify that we can make $\frac{1}{n^p} < \epsilon$ for large n. Doing some rearranging, we want $n > (\frac{1}{\epsilon})^{1/p}$. Here we use two facts: that $0 < x < y$ implies $0 < \frac{1}{y} < \frac{1}{x}$ and that the function $f(x) = x^\alpha$ is monotonically increasing for $\alpha > 0$ (i.e. $x \leqslant y$ implies $x^\alpha \leqslant y^\alpha$).

For (b), we look at three cases: $p > 1$, $p = 1$, and $0 < p < 1$. If $p > 1$, then $\sqrt[n]{p} > 1$ and take $x_n = \sqrt[n]{p} - 1 > 0$. By the binomial theorem:

$$1 + nx_n \leqslant 1 + nx_n + \binom{n}{2} x_n^2 + \cdots = (1 + x_n)^n = p$$

So:

$$0 < x_n \leqslant \frac{p-1}{n}$$

Thus, $x_n \to 0$ so $\sqrt[n]{p} \to 1$. If $p = 1$ the statement is trivial. If $0 < p < 1$ then we take reciprocals and proceed.

For (c), proceed similarly as (b) but let $x_n = \sqrt[n]{n} - 1$. We take a different term in the binomial theorem:

$$\binom{n}{2} x_n^2 \leqslant (1 + x_n)^n = n$$

and find that

$$0 < x_n < \sqrt{\frac{2}{n-1}}$$

so

$$0 < x_{n+1} < \sqrt{2}(\frac{1}{n})^{1/2}$$

and use part (a).

For (d), let k be a positive integer such that $k > \alpha$ (we will need to take binomial terms up to k to dominate α). For $n > 2k$,

$$\frac{n^k p^k}{2^k k!} < \frac{n(n-1)\cdots(n-k+1)}{k!} p^k = \binom{n}{k} p^k \leqslant (1+p)^n$$

and so

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} (\frac{1}{n})^{k-\alpha}$$

and use part (a).

For (e), just take $\alpha = 0$ and $1 + p = x$ from (d). $\square$

9 Lecture 9

9.1 Series

Definition 9.1. Let $\{s_n\}$ be a sequence in $\mathbb{R}$. The *nth partial sum* is

$$S_n := \sum_{i=1}^{n} s_i$$

and we write

$$\sum_{i=1}^{\infty} s_n := \lim_{n\to\infty} S_n$$

to be the *series* of s_n.

Remark 9.2. Much of our study of series will be to find necessary and sufficient conditions for convergence in different situations, based on our knowledge of sequences. This is analogous to studying integrals in terms of functions.

The following is an easy but useful criterion for the convergence of series. It is essentially the Cauchy criterion.

Proposition 9.3. *The series for s_n converges if and only if for every $\epsilon > 0$ there is an integer N such that $m, n > N$ implies that*

$$|\sum_{i=n}^{m} s_i| < \epsilon$$

Remark 9.4. Note that while this seems like an easy result, we had to do a lot of work to prove that Cauchy sequences converge. In general it's harder for us to identify the exact number that a series converges to versus a sequence (maybe in the same way it's hard to take integrals), so the Cauchy criterion is really useful for series.

Corollary 9.5. *If $\sum s_n$ converges, then $s_n \to 0$.*

Proof. Take $n = m$ above. □

Example 9.6 (Counterexample to converse)**.** Consider $s_n = \frac{1}{n}$, which has $\lim_{n\to\infty} s_n = 0$. The corresponding series does not converge. To see why, consider "grouping" the sums as follows:

$$1 + \frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}) + \cdots > 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots$$

Note that in this argument I am not rearranging the series or changing it any way. Rearranging the terms in series must be done delicately, as we will show later.

Proposition 9.7 (Comparison test)**.** *If $|a_n| \leqslant c_n$ for $n > N_0$ (some N_0), and $\sum c_n$ converges, then $\sum a_n$ converges. If $a_n \geqslant d_n \geqslant 0$ for $n > N_0$, and $\sum d_n$ diverges, then $\sum a_n$ diverges.*

Proof. Note that this is a little deeper than a simple application of the comparison of sequences result we proved earlier, since there is no mention of a lower bound converging to the same number. Let $N \geqslant N_0$ be such that $m, n > N$ implies $|\sum_{i=n}^{m} c_i| < \epsilon$. Then we have

$$|\sum_{i=n}^{m} a_i| \leqslant \sum_{i=n}^{m} |a_i| \leqslant \sum_{i=n}^{m} c_i < \epsilon.$$

The second statement follows from comparison of sequences. Note that since all the d_n terms are positive, the partial sums form a monotonically increasing sequence, and since it diverges it must in fact diverge to infinity. Since $a_n \geqslant d_n$, then the series for a_n must also diverge to infinity. □

Example 9.8. Note that it is important for the above c_n to have only positive terms, since must serve as an "overestimate" of a convergent series. For example, if we take $c_n = (-1)^n \frac{1}{n}$, we find that $\sum c_n$ converges, whereas $\frac{1}{n}$ does not. Likewise, it is important for d_n and a_n to be entirely positive or entirely negative.

9.2 Adding, regrouping series

Proposition 9.9. *Let $\sum a_n = A$ and $\sum b_n = B$. Then, $\sum(a_n + b_n) = A + B$ and $\sum ca_n = cA$ for any c.*

Proof. Straightforward, left as exercise. □

Definition 9.10. A *regrouping* of a series $\sum a_n$ is defined as a subsequence of the sequence of partial sums $S_n = \sum_{i=1}^{n} s_i$. This is best exhibited by example. Recall our regrouping of $a_n = \frac{1}{n}$:

$$1 + \frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}) + \cdots > 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots$$

This corresponds to taking the partial sums:

$$S_1, S_2, S_4, S_8, S_{16}, \ldots$$

The way to think of this is that we want to treat the parentheses as one "group" and ignore any intermediate terms one might obtain by only summing some part. Note that this only allows for finitely many terms in each group.

Proposition 9.11. *Let $\sum a_i$ be a series of non-negative terms. Then any regrouping of the series has the same behavior as the series itself.*

Proof. Since the terms are nonnegative, the sequence is monotonically increasing, and thus the set of subsequential limit has at most one point. □

Example 9.12. Let $a_n = (-1)^n$. Note that the following regrouping:

$$(1 + -1) + (1 + -1) + (1 + -1) + \cdots$$

would change the behavior of this series, since it makes this divergent series converge to zero.

Definition 9.13. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a injective and surjective function. Then the *rearrangement* of the sequence $\{a_n\}$ corresponding to σ is given by:

$$(a^\sigma)_k = a_{\sigma(k)}$$

Note that I do not think this notation is standard.

Proposition 9.14. *Suppose σ fixes all but finitely many numbers, i.e. $\sigma(n) = n$ for all but finitely many n. Then, the convergence behavior and also limits of two sequences a^σ and a are the same.*

Proof. This is left as an exercise. □

9.3 "Toolbox" series

Of course, we need some series to compare to.

Proposition 9.15 (Geometric series)**.** *If $x \in [0, 1)$ then*

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

and if $x \geqslant 1$ then the series diverges.

Proof. We can compute by induction the partial sums:

$$\sum_{n=0}^{k} x^n = \frac{1 - x^{k+1}}{1-x}$$

and use previous results on sequences. For $x = 1$ we see immediately it diverges, and we can use the comparison test for other cases. □

Proposition 9.16 (Negative power series)**.** *$\sum_n \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leqslant 1$.*

Proof. We have shown divergence for $p = 1$ and the comparison test takes care of $p \leqslant 1$. For $p > 1$ we will use a similar technique as for $p = 1$, i.e. we will group:

$$1 + (\frac{1}{2^p} + \frac{1}{3^p}) + (\frac{1}{4^p}) + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p}) + (\frac{1}{8^p} + \cdots) + \cdots \leqslant 1 + \frac{1}{2^p} + \frac{2}{2^p} + (\frac{4}{4^p} + \frac{8}{8^p} + \cdots)$$

The term in the parentheses is a geometric series with ratio $\frac{1}{2^{p-1}}$ which is less than 1 if $p - 1 > 0$, and the result follows. □

Proposition 9.17 ($n \log(n)$)**.** *If $p > 1$ then*

$$\sum_{n=2}^{\infty} \frac{1}{n(\log(n))^p}$$

converges. If $p \leqslant 1$ the series diverges.

Proof. This is left as an exercise; use the grouping technique. □

10 Lecture 10

10.1 Root and ratio tests

Theorem 10.1 (Root test)**.** *Consider a sequence a_n and let $\alpha = \limsup \sqrt[n]{|a_n|}$. Then,*
(a) if $\alpha < 1$ then $\sum a_n$ converges,
(b) if $\alpha > 1$ then $\sum a_n$ diverges.

Remark 10.2. Note that if $\alpha = 1$ we cannot conclude anything. For example, $\sum \frac{1}{n}$ diverges, but $\frac{1}{n^2}$ converges, but they both have $\alpha = 1$.

Proof. If $\alpha < 1$, then there is some ϵ such that $\alpha + \epsilon < 1$. By definition of lim sup, there is some N such that $\sup\{\sqrt[N]{|a_N|}, \sqrt[N+1]{|a_{N+1}|}, \ldots\} < \alpha + \epsilon < 1$. In other words, $|a_n| < (\alpha + \epsilon)^n$ for every $n > N$. Thus, $\sum_{n=N}^{\infty} |a_n|$ converges by the comparison test.

If $\alpha > 1$, then take $\epsilon = \frac{\alpha - 1}{2}$ and let $\gamma = \alpha - \epsilon$. There are is an N such that for $n > N$, we have $\sup\{\sqrt[n]{a_n}, \sqrt[n+1]{a_{n+1}}, \ldots\} > \gamma > 1$. Thus we have a subsequence of a_n whose terms are all greater than 1, so the sequence a_n cannot converge to zero.

Note that this argument fails when $\alpha = 1$; in this case we have $\epsilon = 0$. Also note that we cannot use the comparison test and geometric series in analogy to the $\alpha < 1$ case, due to subtle differences in the comparison test for divergence requiring all terms to be positive or all terms to be negative. If $\alpha = 1$, note that $\sum \frac{1}{n}$ diverges but $\sum \frac{1}{n^2}$ converges. Thus the test cannot be conclusive in this case. □

Theorem 10.3 (Ratio test)**.** *The series $\sum a_n$:*
(a) converges if $\limsup |\frac{a_{n+1}}{a_n}| < 1$
(b) diverges if $|\frac{a_{n+1}}{a_n}| \geqslant 1$ for all $n > N_0$ for some fixed N_0

Proof. For (a), fix some $\beta \in (\limsup |\frac{a_{n+1}}{a_n}|, 1)$; there is some N such that if $n \geqslant N$ we have that $|\frac{a_{n+1}}{a_n}| < \beta$. Thus, we find that:

$$|a_{N+1}| < \beta |a_N|$$

and inductively we find

$$a_{N+k}| < \beta^k |a_N|$$

i.e.

$$|a_n| < \beta^{n-N} |a_N|$$

for $n > N$. Since $\sum \beta^n$ converges, by the comparison test, $\sum a_n$ converges. For (b), check that a_n cannot limit to zero. □

Remark 10.4 (Similarities and differences)**.** The techniques for both tests are relatively similar: for convergence we use comparison with a geometric series, and for divergence we show that the sequence does not limit to zero. However, there is a difference in divergence for the ratio test, which insists that the ratio must be greater than or equal to one for *all terms* past a point, whereas the ratio test asks for a limit supremum to be positive. If we try to apply the ratio test to the series $\sum \frac{1}{n}$, we find that it falls under neither (a) nor (b), since the limit supremum of $|\frac{n}{n+1}|$ is 1 but is never greater than 1.

Example 10.5. This example will show that the divergence part of the ratio test is extremely blunt. Take the series:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \cdots$$

and note that (index $a_2 = \frac{1}{2}$):

$$\limsup \sqrt[n]{a_n} = \lim \sqrt[2n]{\frac{1}{2^n}} = \frac{1}{\sqrt{2}}$$

whereas

$$\limsup |\frac{a_{n+1}}{a_n}| = \infty$$

since we always have consecutive terms of the form $\frac{1}{2^k}, \frac{1}{3^{k+1}}$.

Example 10.6. This example is a "rearrangement" of a convergent series

$$\frac{1}{2} + 1 + \frac{1}{8} + \frac{1}{4} + \frac{1}{32} + \frac{1}{16} + \cdots$$

obtained by swapping pairs of terms. Notice that:

$$\limsup \sqrt[n]{a_n} = \frac{1}{2}$$

but

$$\limsup |\frac{a_n}{a_{n+1}}| = 2$$

Notice that:

$$\liminf |\frac{a_n}{a_{n+1}}| = \frac{1}{8}$$

and that the geometric mean of 2 and $\frac{1}{8}$ is $\frac{1}{2}$. In some really vague sense, the ratio test is too sensitive to the ordering of the sequence.

Proposition 10.7 (The root test is stronger)**.** *For any sequnece a_n of positive numbers,*

$$\limsup \sqrt[n]{a_n} \leqslant \limsup \frac{a_{n+1}}{a_n}$$

Proof. Choose some $\beta > \limsup \frac{c_{n+1}}{c_n}$. There is an N such that $n \geqslant N$ implies that $\frac{a_{n+1}}{a_n} < \beta$. Thus one finds that $a_n < \beta^{n-N} a_N$. Taking nth roots, one obtains that $\sqrt[n]{a_n} < \beta \sqrt[n]{\beta^N a_N}$. Note that on the right hand side $\beta^N a_N$ is a constant, and taking lim sup we find that $\limsup \sqrt[n]{a_n} < \beta$. Since this is true for every $\beta > \limsup \frac{a_{n+1}}{a_n}$, one obtains the result. □

10.2 Summation by parts, alternating series

Theorem 10.8 (Summation by parts)**.** *Let a_n and b_n be two sequences, and write $A_n = sum_{i=0}^n a_i$ for $n \geqslant 0$ and $A_{-1} = 0$. Then for $0 \leqslant m \leqslant n$ one has:*

$$\sum_{i=m}^{n} a_n b_n = \sum_{i=m}^{n-1} A_i(b_i - b_{i+1}) + A_n b_n - A_{m-1} b_m$$

Proof. Write $a_n = A_{n+1} - A_n$. □

Proposition 10.9. *If,*
(a) the partial sums A_n of a_n are bounded,
(b) b_n is monotonically decreasing,
(c) b_n converges to zero,
then $\sum a_n b_n$ converges.

Proof. Suppose $|A_n| < M$, i.e. M is a bound. We want to use the Cauchy criterion, i.e. we want to make the following expression $< \epsilon$:

$$|\sum_{i=m}^{n} a_n b_n| = |\sum_{i=m}^{n-1} A_i(b_i - b_{i+1}) + A_n b_n - A_{m-1} b_m| < M(\sum_{i=m}^{n-1} |b_i - b_{i+1}| + |b_n| + |b_m|)$$

Since $b_i - b_{i+1} \geqslant 0$ and $b_i \geqslant 0$, this is:

$$M(\sum_{i=m}^{n-1} b_i - b_{i+1} + b_n + b_m) = 2Mb_m$$

Now, by convergence of b_n, for any $\epsilon > 0$, we can find an N such that $m > N$ implies that $b_m < \frac{\epsilon}{2M}$, completing the proof. $\square$

Proposition 10.10 (Alternating series test)**.** *A sequence is* alternating *if its signs alternate, i.e. if $a_i \geqslant 0$ then $a_{i+1} \leqslant 0$ and $a_{i+2} \geqslant 0$, et cetera. Let a_n be an alternating series such that $|a_n|$ is monotonically decreasing and a_n converges to 0. Then, $\sum a_n$ converges.*

Proof. Apply the previous proposition with $a_n = (-1)^n$ and $b_n = |c_n|$. $\square$

Example 10.11 (Why monotonically decreasing)**.** Otherwise, one could make the negative terms converge much faster than the positive terms. For example, take $a_{2n} = \frac{1}{n}$ and $a_{2n+1} = -\frac{1}{2^n}$. One can check by regrouping:

$$1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{4} + (\frac{1}{3} - \frac{1}{8} + \frac{1}{4} - \frac{1}{16}) + \cdots$$

that the groups are all $\geqslant \frac{1}{4}$.

10.3 Absolute convergence, multiplying and rearranging series

Definition 10.12. A series $\sum a_n$ converges *absolutely* if $\sum |a_n|$ converges. It is easy to see an absolutely convergent series is convergent. A series that converges but not absolutely is called *conditionally* convergent.

Remark 10.13. The root test and ratio tests all use $|a_n|$ rather than a_n, and thus are really testing for absolute convergence. Summation by parts does handle some non-absolutely convergent series.

Example 10.14 (A conditionally convergent series)**.** By the alternating series test, $\sum(-1)^n \frac{1}{n}$ converges, but $\sum \frac{1}{n}$ does not.

Definition 10.15. For series $\sum a_n$ and $\sum b_n$, define their product, which we will write by informal convention as $\sum c_n$, by

$$c_n = \sum_{i+j=n} a_i b_j = \sum_{k=0}^{n} a_k b_{n-k}$$

This definition is motivated by imagining products of power series in the formal variasble x:

$$(\sum a_n x^n)(\sum b_n x^n) = \sum_{n=1}^{\infty} \sum_{i+j=n} a_i b_j x^n$$

Remark 10.16. Notice that we can't apply sequence techniques to conclude that $AB = C$ (the infinite sums), since $A_n B_n \neq C_n$ (the partial sums).

Example 10.17. Take $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$. These both converge by the alternating series test. However, the product has:

$$c_n = (-1)^n \sum_{i+j=n} \frac{1}{\sqrt{ij}}.$$

Now, notice that ij is maximized when $i = j = \frac{n}{2}$. Thus we have that

$$c_n \geqslant \frac{n-2}{\sqrt{n^2/4}} = 2 - \frac{4}{n}$$

and thus c_n does not converge to 0, so the series cannot converge.

Proposition 10.18. *Using the previous notation, if one of $\sum a_n, \sum b_n$ converges absolutely, then $AB = C$.*

Proof. For shorthand, write $\beta_n = B_n - B$. We can write

$$C_n = a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 = a_0(B + \beta_n) + a_1(B + \beta_{n-1}) + \cdots + a_n(B + \beta_0)$$

$$= A_n B + a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$$

To prove the result, we need to show that the "tail:"

$$\gamma_n = \sum_{i+j=n} a_i\beta_j = a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$$

converges to zero.

Since $\sum a_n$ converges absolutely, let $\alpha = \sum |a_n|$. Since β converges to 0, let N be such that $n \geqslant N$ implies that $\beta_n < \epsilon$. Then, we can break up the sum into two groups using the triangle inequality:

$$|\gamma_n| \leqslant |\beta_0\alpha_n + \cdots + \beta_N a_{n-N}| + |\beta_{N+1}a_{n-N-1} + \cdots + \beta_n a_0|$$

$$\leqslant |\beta_0\alpha_n + \cdots + \beta_N a_{n-N}| + \alpha\epsilon$$

Taking lim sup of both sides, we find that $\limsup |\gamma_n| \leqslant \epsilon\alpha$ (we can't take the limit because we don't know it exists). Since ϵ was chosen arbitrarily, it must follow that $\limsup \gamma_n = 0$, and the result follows. □

Definition 10.19. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a injective and surjective function. Then the *rearrangement* of the sequence $\{a_n\}$ corresponding to σ is given by:

$$(a^\sigma)_k = a_{\sigma(k)}$$

Note that I do not think this notation is standard.

Example 10.20. Take $a_n = \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$. The series converges by the alternating series test, to a number $A < 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$.

Now, we can rearrange this series so that it has two positive numbers followed by one negative:

$$[1 + \frac{1}{3} - \frac{1}{2}] + [\frac{1}{5} + \frac{1}{7} - \frac{1}{4}] + [\frac{1}{9} + \frac{1}{11} - \frac{1}{6}]\cdots$$

(the brackets are only for visual ease). Each three term group is:

$$\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k}$$

and one can verify that this is positive when $k \geqslant \frac{3}{8}$. This the partial sums are monotonically increasing after a point, and one can check that eventually the sum becomes greater than $\frac{5}{6}$. Thus the two cannot converge to the same number.

Proposition 10.21. *Suppose σ fixes all but finitely many numbers, i.e. $\sigma(n) = n$ for all but finitely many n. Then, the convergence behavior and also limits of two sequences a^σ and a are the same.*

Proof. This is left as an exercise. □

Proposition 10.22. *If $\sum a_n$ converges absoutely, then every rearrangement converges to the same number.*

Proof. Let $\sum a'_n$ be the rearrangement. For $\epsilon > 0$, choose N such that for $m, n > N$, one has that $\sum_{i=m}^n |a_i| < \epsilon$ and $\sum_{i=m}^n |a'_i| < \epsilon$. So, we need to "take care" of terms $a_1, \ldots, a_N$.

Now, choose M such that $a'_1, \ldots, a'_M$ contains all the $a_1, \ldots, a_N$ terms. Then, in the quantity $|A_n - A'_n|$, the first N of the original series will cancel in both A_n and A'_n. Using the triangle inequality, this quantity is $< 2\epsilon$. □

Proposition 10.23. *Suppose $\sum a_n$ converges conditionally. Let $\alpha \leqslant \beta$. Then there is a rearrangement of a_n which has $\liminf = \alpha$ and $\limsup = \beta$. One can take α, β to be infinite.*

Proof. I will skip this. Refer to Rudin. □

11 Lecture 11

11.1 Limits of functions

Definition 11.1. Let X, Y be metric spaces, and $E \subset X$, and p a limit point of E. We write the *limit*

$$\lim_{x \to p} f(x) = q$$

if for every $\epsilon > 0$ there exists a $\delta > 0$ such that $x \in B_\delta(p) - \{p\}$ implies that $f(x) \in B_\epsilon(q)$. Note that it is not relevant whether or not $p \in E$. Also, I will use the same symbols for this set-up throughout this section.

Example 11.2 (Sequences)**.** We can realize $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$ as the subset $\{\frac{1}{n} \mid n \in \mathbb{N}\} \cup \{0\} \subset \mathbb{R}$. Here, ∞ is the only limit point, and the definitions agree.

Example 11.3 (One-sided limits in $\mathbb{R}$)**.** In calculus sometimes we take one-sided limits, i.e. $\lim_{x \to 0^+} \lfloor x \rfloor = 0$ but $\lim_{x \to 0^-} \lfloor x \rfloor = -1$. Here, the subset $E = (0, \infty)$ and $(-\infty, 0)$ respectively.

Proposition 11.4. *One has $\lim_{x \to p} f(x) = q$ if and only if for every sequence $\{p_n\}$ such that $p_n \neq p$ (for all n) and $p_n \to p$, one has $\lim_{n \to \infty} f(p_n) = q$.*

Proof. Suppose that $\lim_{x \to p} f(x) = q$. Then, for every $\epsilon > 0$ one can find a $\delta > 0$ such that $x \in B_\delta(p) - \{p\}$ implies that $f(x) \in B_\epsilon(q)$. Any sequence p_n converging to p and such that $p_n \neq p$ has an N such that $p_n \in B_\delta(p) - \{p\}$ for $n > N$, and so $f(p_n) \in B_\epsilon(q)$. Thus we have shown that for every ϵ there is an N such that $f(p_n) \in B_\epsilon(q)$.

If one did not have that $\lim_{x \to p} f(x) = q$, then there is some $\epsilon > 0$ for which one could find a p_n arbitrarily close to p (i.e. for every δ) such that $f(p_n) \notin B_\epsilon(q)$. Fix this ϵ; choose $p_n \in B_{\frac{1}{n}}(p) - \{p\}$ such that $f(p_n) \notin B_\epsilon(q)$. This is a sequence such that $f(p_n)$ does not converge to q. □

Corollary 11.5. *Limits of functions are unique, if they exist.*

Proposition 11.6. *Let $\lim_{x \to p} f(x) = A$ and $\lim_{x \to p} g(x) = B$. Then,*
(a) $\lim (f + g)(x) = A + B$,
(b) $\lim (fg)(x) = AB$,
(c) $\lim \frac{f}{g}(x) = A/B$ *if* $B \neq 0$

Proof. Follows from the result in sequences and the previous proposition. □

Example 11.7. We can check explicitly: $\lim_{x \to a} x = a$. Fix an $\epsilon > 0$; we are concerned with the interval $(a - \epsilon, a + \epsilon)$. We want to show there is δ such that $x \in (a - \delta, a + \delta)$ implies that $f(x) \in (a - \epsilon, a + \epsilon)$. But we can just take $\delta := \epsilon$!

Example 11.8. Define $f(x)$ to be 0 if $x \in \mathbb{Q}$ and 1 if $x \notin \mathbb{Q}$. The limit $\lim_{x \to 0} f(x)$ does not exist, since in every interval $(-\epsilon, \epsilon)$ there are points such that $|f(x) - f(0)| = 1$ and $|f(x) - f(0)| = 0$.

11.2 Continuity

Definition 11.9. Let $f : X \to Y$ be a function, and $U \subset Y$ a subset. Then define the *inverse image*:

$$f^{-1}(U) := \{x \in X \mid f(x) \in U\}$$

Let $V \subset X$ be a subset, and define the *image*

$$f(V) := \{f(x) \in Y \mid x \in X\}$$

Definition 11.10. Let X and Y be metric spaces. A function $f : X \to Y$ is *continuous* at $p \in X$ if the following equivalent are true:
(a) $\lim_{x \to p} f(x) = f(p)$,
(b) for every $\epsilon > 0$, there is a $\delta > 0$ such that $f(x) \in B_\epsilon(f(p))$ for all $x \in B_\delta(p)$,
(c) for every $\epsilon > 0$, there is a $\delta > 0$ such that $f(B_\delta(p)) \subset B_\epsilon(f(p))$.

Definition 11.11. Let X and Y be metric spaces. A function $f : X \to Y$ is *continuous* if the following equivalent are true:
(a) for every $p \in X$, $\lim_{x\to p} f(x) = f(p)$,
(b) for every $p \in X$ and every $\epsilon > 0$, there is a $\delta > 0$ such that $f(x) \in B_\epsilon(f(p))$ for all $x \in B_\delta(p)$,
(c) for every open subset $U \subset Y$, $f^{-1}(U)$ is open,
(d) for every closed subset $Z \subset Y$, $f^{-1}(Z)$ is closed.

Proof. The equivalence of (c) and (d) is a set-theoretic exercise, knowing that U is open if and only if U^c is closed. The equivalence of (a) and (b) is simply untangling the definition. I will show that (b) and (c) are equivalent. Assume (a). Let $U \subset Y$ be open, and let $p \in f^{-1}(U)$. We want to show that p is an interior point. Since $f(p) \in U$ it is an interior point, so choose ϵ such that $B_\epsilon(f(p)) \subset U$. Then there is a δ such that $f(B_\delta(p)) \subset B_\epsilon(f(p)) \subset U$ and thus $B_\delta(p) \subset f^{-1}(U)$. Now assume (c). Then, $f^{-1}(B_\epsilon(f(p))$ is open, so p is an interior point, so there is some $B_\delta(p) \subset f^{-1}(B_\epsilon(f(p))$. □

Example 11.12. In the discrete metric, all functions are continuous.

Remark 11.13. Note that it is not true that if f is continuous at x, then it is continuous in a neighborhood of x. As a counterexample, consider the function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 0$ if $x \notin \mathbb{Q}$ and $f(x) = x$ if $x \in \mathbb{Q}$. Check that f is continuous only at 0.

Proposition 11.14 (Composition of continuous functions)**.** *Let $f : X \to Y$ and $g : Y \to Z$. If f is continuous at $p \in X$ and g is continuous at $f(p) \in Y$, then $g \circ f$ is continuous at p.*

Proof. This proof is just unwinding a lot of definitions. For any $\epsilon > 0$ we want to show that there is a $\delta > 0$ such that $g(f(B_\delta(p)) \subset B_\epsilon(g(f(p))$. We know that we can find a $\delta' > 0$ such that $g(B_{\delta'}(f(p)) \subset B_\epsilon(g(f(p))$. Using δ' as the "ϵ" for f, we then find a δ such that $f(B_\delta(p)) \subset B_{\delta'}(f(p))$. □

Corollary 11.15. *Let $f : X \to Y$ and $E \subset X$. Suppose $L = \lim_{x\to a} f(x)$ exists. If $g : f(X) \cup \{L\} \to Z$ is continuous at L, then*

$$\lim_{x\to a} g(f(x)) = g(L).$$

Proposition 11.16 (Algebraic operations on continuous functions in $\mathbb{R}$)**.** *If f and g are continuous then $f + g, fg, f/g$ are continuous (assume $g(x) \neq 0$ for the last one).*

12 Lecture 12

12.1 Properties of continuity

Proposition 12.1. *Let X be compact, and $f : X \to Y$ a continuous function. Then the image, $f(X)$, is compact.*

Proof. Take an open cover of $f(X)$ and pull it back using f^{-1}. Since f is continuous, this is an open cover of X. It has a finite subcover since X is compact. The corresponding subcollection in the original cover of $f(X)$ is thus a subcover. □

Remark 12.2. Note that it is not true that $f(U)$ is open for U open. For example, let $f : \mathbb{R} \to \mathbb{R}^2$ send $x \mapsto (x, 0)$. Let U be any open interval; the image is not open in $\mathbb{R}^2$. It is also not true that $f(Z)$ is closed for Z closed. For example, let $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = e^x$, and consider the image $f((-\infty, a]) = (0, e^a]$ for any a. It is also not true that $f(E)$ is bounded for E bounded. Let $f : \mathbb{R} - \{0\} \to \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$, and consider $f((0, 1)) = (1, \infty)$. Nor is it true that $f^{-1}(E)$ is bounded for E bounded. Let $f : \mathbb{R}\mathbb{R}$ be a constant function, say $f(x) = 0$. Then $f^{-1}(\{0\}) = \mathbb{R}$ is not bounded.

Theorem 12.3 (Maximum value theorem)**.** *Let $[a, b]$ be any nonempty closed interval. Suppose that $f : [a, b] \to \mathbb{R}$ is a continuous function of real numbers. Then, f attains maximum and minimum values on $[a, b]$.*

Proof. This follows immediately; $f([a, b])$ is closed and bounded. □

In the homework, the student explored connectedness. Recall the definition.

Definition 12.4. Two subsets of X are *separated* if $A \cap \overline{B} = \overline{A} \cap B = \varnothing$. A subset $E \subset X$ is *connected* if it is not the union of two nonempty separated sets.

Proposition 12.5. *A subset $\mathbb{E}$ of $\mathbb{R}$ is connected if and only if for all pairs, $x, y \in E$, one has $(x, y) \subset E$.*

Proof. If the condition is false then one can take $z \in (x, y)$ and one has $(-\infty, z)$ and (z, ∞) separating E. Conversely, if E is not connected, take nonempty separated sets A and B covering E. Choose $x \in A$ and $y \in B$ and take $z = \sup A \cap [x, y]$. Since $z \in \overline{A}$, $z \neq B$. If $z \notin A$ then $z \in (x, y)$ breaks the condition. If $z \in A$ then $z \notin \overline{B}$, so one can find $z' \in (z, y)$ such that $z' \notin B$, but also $z' \notin A$, so $z' \notin E$, breaking the condition. □

Proposition 12.6. *Let $E \subset X$ be connected, and $f : X \to Y$ continuous. Then $f(E)$ is connected.*

Proof. Suppose $f(E)$ is not connected; then it has separating sets A and B. We claim that $f^{-1}(A)$ and $f^{-1}(B)$ are separating sets for E. They are clearly nonempty and cover E. Using two facts: (1) $f^{-1}(\overline{A}) \subset \overline{f^{-1}(A)}$ and (2) $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$, one has that $\overline{f^{-1}(A)} \cap f^{-1}(B) = f^{-1}(\overline{A}) \cap f^{-1}(B) = f^{-1}(\overline{A} \cap B) = \varnothing$. Check the two facts as an exercise. □

Theorem 12.7 (Intermediate value theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. If $f(a) < f(b)$, and $c \in (f(a), f(b))$, then there is some $x \in (a, b)$ such that $f(x) = c$.*

Proof. Apply the previous two propositions. □

Corollary 12.8 (Fixed point theorem for $\mathbb{R}$)**.** *Let $f : [0, 1] \to [0, 1]$ be a continuous map. Then f has a fixed point, i.e. some p such that $f(p) = p$.*

Proof. Apply the IVT to $g(x) = f(x) - x$. Note that $g(0) = f(0) \geqslant 0$ and $g(1) = f(1) - 1 \leqslant 0$, one has some p such that $g(p) = 0$. □

Proposition 12.9. *Let f be a one-to-one continuous function on an interval I, i.e. $f : I \to \mathbb{R}$. Then f is either strictly increasing or strictly decreasing. Recall that $f : I \to \mathbb{R}$ is* strictly increasing *if for all $a < b \in I$, one has that $f(a) < f(b)$.*

Proof. First, we will show the result for a closed interval $I = [a, b]$. Suppose that $f(a) < f(b)$ (for $f(b) < f(a)$ the proof is the same). We want to show that every $c \in (a, b)$ has $f(c) \in (f(a), f(b))$. If not, then if $f(c) < f(a)$, then $f(a) \in (f(c), f(b))$, so there is some $p \in (c, b)$ such that $f(p) = f(a)$ by the IVT, contradicting injectivity. If $f(c) > f(b)$ a similar argument applies. Thus, $f(c) \in (f(a), f(b))$.

Now, if the interval were open, say $I = (a, b)$, then one could take a closed subinterval $[a', b'] \subset (a, b)$ and extend the argument as follows. Let $c \in (a, a']$. We wish to show that $f(c) < f(a')$. If not, i.e. $f(c) > f(a')$, then choose p such that $f(p) \in (f(a'), f(c))$. Then apply the IMT to contradict injectivity, and this completes the proof. □

13 Lecture 13

13.1 Uniform continuity

Definition 13.1. Let $f : X \to Y$ be a function on metric spaces. We say that f is *uniformly continuous* on X if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$d(f(x), f(x')) < \epsilon$$

for all x, x' such that $d(x, x') < \delta$.

Remark 13.2. Unlike the notion of continuity, which only had a topological characterization (i.e. in terms of open sets only), uniform continuity requires the use of a metric. There are ways around this (e.g. uniform spaces) but we won't get into that here.

Example 13.3. Consider $f(x) = \frac{1}{x}$. We claim that $f(x)$ is uniformly continuous on $[a, \infty)$ for any $a > 0$. Fix $\epsilon > 0$; one can show because $f(x)$ is monotonically decreasing, that for fixed x, $d(f(x), f(x'))$ is maximized when $x' = x + \delta$ or $x' = x - \delta$. Then,

$$|f(x) - f(x+\delta)| = |\frac{1}{x} - \frac{1}{x+\delta}| = \frac{\delta}{x(x+\delta)}.$$

This is maximized when x is minimal, i.e. at $x = a$, and takes value $\frac{\delta}{a^2+\delta a}$. Thus, for $\epsilon > 0$ we want to find a δ such that

$$\frac{\delta}{a^2 + \delta a} < \epsilon$$

Check that

$$\delta < \frac{\epsilon a^2}{1 - a\epsilon}$$

are all good choices. One has to check that the right hand side is positive, which is true as long as $\epsilon < \frac{1}{a}$. However for larger ϵ check that in fact any δ will do.

However, $f(x) = \frac{1}{x}$ is not uniformly continuous on the interval $(0, \infty)$. Fix $\epsilon > 0$. I claim that we can find two x, x' arbitrarily close to each other such that $d(f(x), f(x')) > \epsilon$. Note that for $x = \frac{1}{n}$ and $x' = \frac{1}{n+1}$ one has that $|f(x) - f(x')| = n$. For n large enough, one can make $|x - x'|$ arbitrarily small, and at the same time make n arbitrarily large.

Proposition 13.4. *Let X be compact and $f : X \to Y$ be continuous. Then f is uniformly continuous.*

Proof. Fix $\epsilon > 0$. By continuity of f, for each $x \in X$ choose $\delta_x > 0$ such that $f(B_{\delta_x}(x)) \subset B_{\frac{\epsilon}{2}}(f(x))$. Consider the open cover:

$$\{J_x := B_{\frac{\delta_x}{2}}(x)\}_{x \in X}$$

This has a finite subcover by compactness. Let that finite subcover by indexed by I; then take $\delta = \frac{1}{2}\min(\delta_x \mid x \in I)$.

Now, we claim that for $p, q \in X$, that $d(p, q) < \delta$ implies that $d(f(p), f(q)) < \epsilon$. Suppose p be in the open given by $i \in I$ and q in the open given by $j \in I$. Since $d(p, q) < \delta$, there must be a point in common between the balls J_i and J_j, call it x. Then,

$$d(f(p), f(q)) \leqslant d(f(p), f(x)) + d(f(x), f(q)) < \epsilon$$

as desired. □

Proposition 13.5. *Let $f : X \to Y$ be uniformly continuous. Let s_n be a Cauchy sequence in X. Then $f(s_n)$ is a Cauchy sequence in Y.*

Remark 13.6. Note that this is untrue for continuous functions. For continuous functions, we insist that s_n is convergent.

Proof. Fix $\epsilon > 0$. We want to find an N such that $m, n > N$ implies that $|f(s_n) - f(s_m)| < \epsilon$. By uniform continuity there is a δ such that $|s_n - s_m| < \delta$ implies this, and by Cauchy-ness of s_n there is an N that guarantees this. □

13.2 The derivative

Definition 13.7. Let $f : [a, b] \to \mathbb{R}$ be a real valued function. Define the *derivative*:

$$f'(x) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x}$$

Note that this limit may not always be well-defined at every point. If $f'(x)$ exists at x we say f is *differentiable* at x.

Proposition 13.8. *If f is differentiable at x, then f is continuous at x.*

Proof. Using limit laws,

$$\lim_{t\to 0} f(x) - f(t) = \lim_{t\to 0} \frac{f(x)-f(t)}{x-t}(x-t) = f'(x)\cdot 0 = 0$$

□

Remark 13.9. Note that it is not true that if f is differentiable at x, then f is continuous in a neighborhood of x. For example, take

$$f(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ -x^2 & x \notin \mathbb{Q} \end{cases}$$

This function is continuous only at zero, and differentiable only at zero.

Example 13.10. Take

$$f(x) = \begin{cases} x\sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

We claim that the derivative does not exist at 0. To see this, note that

$$\frac{f(x)-f(0)}{x} = \sin(\frac{1}{x})$$

But this has no limit as $x \to 0$.

Example 13.11. Now take

$$f(x) = \begin{cases} x^2\sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

We claim that $f'(0) = 0$.

Proposition 13.12 (Arithmetic operations and the derivative)**.** *Let $f, g : [a,b] \to \mathbb{R}$ and suppose f, g are differentiable at $x \in [a,b]$. Then the following are differentiable at x and are given by:*
(a) $(f+g)'(x) = f'(x) + g'(x)$
(b) $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$
(c) $(f/g)'(x) = \frac{f'(x)g(x)-f(x)g'(x)}{g(x)^2}$

Proof. It is easy to prove (a) from the definition. We will prove (b).

$$\frac{f(t)g(t)-f(x)g(x)}{t-x} = \frac{f(t)[g(t)-g(x)]+g(x)[f(t)-f(x)]}{t-x} = f(t)\frac{g(t)-g(x)}{t-x} + g(x)\frac{f(t)-f(x)}{t-x}$$

Taking the limit as $t \to x$, one has the result. See the textbook for the proof of (c). □

Example 13.13. This allows us to differentiate all polynomials. It is easy to show directly that if $f(x) = 1$ then $f'(x) = 0$. It is also easy to show that if $g(x) = x$ then $g'(x) = 1$. Using the product rule, one has that $(x^n)' = x^{n-1}(x)' + (x^{n-1})'x = x^{n-1} + (x^{n-1})'x$, and one can use an argument by induction to show that $(x^n)' = nx^{n-1}$. Then one uses scaling and addition to get all polynomials.

Proposition 13.14 (Chain rule)**.** *Let $f : [a,b] \to \mathbb{R}$ be differentiable at $x \in [a,b]$, and $g : I \to \mathbb{R}$ where $f([a,b]) \subset I$, and g is differentiable at $f(x)$. Then*

$$(g \circ f)'(x) = g'(f(x))f'(x)$$

Remark 13.15. What one wants to do is write

$$\frac{g(f(t))-g(f(x))}{t-x} = \frac{g(f(t))-g(f(x))}{f(t)-f(x)}\frac{f(t)-f(x)}{t-x}$$

and take the limit. The problem here is we cannot guarantee that $f(t) - f(x) \neq 0$ near $t = x$. For example, considering $f(x) = x^2\sin(\frac{1}{x})$ for $x \neq 0$ and $f(0) = 0$, one that for $x \in B_\delta(0)$ for any δ, f has a zero. So there are functions for which one cannot avoid this problem by taking a sufficiently small δ.

Proof. Put $y = f(x)$. We can find functions u and v such that:

$$f(t) - f(x) = (t - x)(f'(x) + u(t))$$

$$g(s) - g(y) = (s - y)(g'(y) + v(s))$$

and $u(x) = 0$ and $v(y) = 0$. Note that u and v are continuous at x and y respectively, as

$$lim_{t \to 0} \frac{f(t) - f(x)}{t - x} - f'(x) = 0$$

and likewise for v.

Then,

$$g(f(t)) - g(f(x)) = (f(t) - f(x))(g'(f(x)) + v(f(t)) = (t - x)(f'(x) + u(t))(g'(f(x)) + v(f(t)))$$

and for $t \neq x$, one has

$$\frac{g(f(t)) - g(f(x))}{t - x} = (f'(x) + u(t))(g'(f(x)) + v(f(t)))$$

Taking the limit $t \to x$, one finds has the result. □

14 Lecture 14

14.1 Mean value theorem

Definition 14.1. A a point p is a *local maximum* of f if there is a δ such that $f(x) = \max(f(B_\delta(x)))$. Likewise, for local minimum.

Proposition 14.2. *Let $f : [a, b] \to \mathbb{R}$. If f has a local maximum or minimum at p, and if $f(p)$ exists, then $f'(p) = 0$.*

Proof. Suppose f has a local maximum at p. Then we have some $\delta > 0$ such that $f(p) \geqslant f(x)$ for $x \in (p - \delta, p + \delta)$. The derivative is

$$\lim_{t \to p} \frac{f(t) - f(p)}{t - p}$$

Note that this expression is $\geqslant 0$ when $t \leqslant p$ and $\leqslant 0$ when $t \geqslant p$. Thus if this limit exists it must be zero. □

Remark 14.3. One consequence of this proposition is that to find the local minima or maxima of a function, one only needs to look at points where $f'(x) = 0$ or does not exist.

Theorem 14.4 (Rolle's theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous, and suppose f is differentiable on (a, b), and that $f(a) = f(b)$. Then there is some $c \in (a, b)$ such that $f'(c) = 0$.*

Proof. By the maximum value theorem, there is a global maximum on this interval, which must be a local maximum as well, and the result follows from the previous proposition. □

Theorem 14.5 (Generalized mean value theorem)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be continuous and differentiable in (a, b). Then there is $c \in (a, b)$ such that*

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(x)$$

Remark 14.6. One way to think of this is to rearrange terms:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x)}{g'(x)}$$

Proof. Take $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. We want to show that $h'(x) = 0$ for some $c \in (a, b)$. Note that $h(a) = h(b)$ and apply the previous theorem. □

Theorem 14.7 (Mean value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b), then there is some $c \in (a, b)$ such that*

$$f(b) - f(a) = (b - a)f'(c)$$

Proof. Take $g(x) = x$ above. □

Here is an application

Proposition 14.8. *Let f be differentiable on (a, b). Then,*
(a) if f is strictly increasing then $f'(x) > 0$ for $x \in (a, b)$,
(b) if f is strictly decreasing then $f'(x) < 0$ for $x \in (a, b)$,
(c) if f is increasing then $f'(x) \geqslant 0$ for $x \in (a, b)$,
(d) if f is decreasing then $f'(x) \leqslant 0$ for $x \in (a, b)$.

Proof. Let $a < x < y < b$. Then by the mean value theorem,

$$\frac{f(y) - f(x)}{y - x} = f'(c) > 0$$

for some $c \in (x, y)$. Thus $f(y) > f(x)$. □

Proposition 14.9 (Intermediate value theorem for derivatives)**.** *Let $f : [a, b] \to \mathbb{R}$ be differentaible, such that $f'(a) < f'(b)$. Let $\lambda \in (f'(a), f'(b))$. Then there is some $x \in (a, b)$ such that $f'(x) = \lambda$.*

Remark 14.10. Note that this isn't an immediate application of the intermediate value theorem, since derivatives of continuous functions need not be continuous.

Proof. Take $g(x) = f(x) - \lambda x$. We want to show that g has a zero in (a, b). Note that $g'(a) < 0$ and $g'(b) > 0$. Using the definition of derivative, this means there is some point t_1 such that $g(t_1) < g(a)$ and a point t_2 such that $g(t_2) < g(b)$. Thus neither a nor b can be global minimums. Since f is continuous in $[a, b]$, it attains a global in this interval not at the endpoints, and at this point the derivative of g vanishes. □

15 Lecture 15

15.1 L'Hospital's Rule

Theorem 15.1 (L'Hospital's Rule)**.** *Suppose $f, g : [a, b] \to \mathbb{R}$ are differentiable in (a, b). Allow $\pm\infty$ for the symbols a, b. Suppose:*

$$\lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L$$

If one of the following are true:
(a) $\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0$
(b) $\lim_{x \to a^+} g(x) = \infty$
Then one has:

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L$$

The symmetric statements for b^- are also true.

Proof. In this proof we will have to choose a lot of symbols. One should think of c, c', c'' as being increasingly close to a as the number of $'$ symbols increase.

Case 1: A is finite or $A = -\infty$. The assumptions of this case allow us to choose some $q \in (A, \infty)$. **The goal** is to show that there is some interval $(a, a + \delta)$ such that if $x \in (a, a + \delta)$ then $\frac{f(x)}{g(x)} < q$. For technical reasons we will see later, we want to again choose an $r \in (A, q)$, and we will show that $\frac{f(x)}{g(x)} \leqslant r < q$ instead.

Since the limit of the expression is A, there is some c such that $x \in (a, c)$ implies that

$$\frac{f'(x)}{g'(x)} < r$$

Now choose $\alpha < \beta \in (a, c)$. Using the generalized mean value theorem, there is some $t \in (\alpha, \beta)$ such that

$$\frac{f(\alpha) - f(\beta)}{g(\alpha) - g(\beta)} = \frac{f'(t)}{g'(t)} < r$$

If (a) holds, we can take the limit $\alpha \to a$ to find that for $\beta \in (a, c)$, one has

$$\frac{f(\beta)}{g(\beta)} \leqslant r < q$$

This proves the goal in this case.

If (b) holds, we can find some $c' \in (a, c)$ such that $g((a, c')) > \max(0, g(\beta))$, i.e. such that

$$\frac{g(\alpha) - g(\beta)}{g(\alpha)} > 0$$

for $\alpha \in (a, c')$. Multiplying by this factor, one has:

$$\frac{f(\alpha) - f(\beta)}{g(\alpha)} < r\frac{(g(\alpha) - g(\beta)}{g(\alpha)}$$

so

$$\frac{f(\alpha)}{g(\alpha)} < r - r\frac{g(\beta)}{g(\alpha)} + \frac{f(\beta)}{g(\alpha)} = r - \frac{M}{g(\alpha)}$$

for some constant M (i.e. we will fix β and r). Let $\epsilon = \frac{1}{2}(q - r)$. Since $g(\alpha) \to \infty$ as $\alpha \to a$, there is some $c'' \in (a, c)$ such that for $\alpha \in (a, c'')$ one has that $|\frac{M}{g(\alpha)}| < \epsilon$ so that

$$\frac{f(\alpha)}{g(\alpha)} < r + \epsilon < q$$

This proves the goal in this case.

Case 2 A is finite or $A = \infty$. One can repeat the argument for $q < A$. Together, this completes the argument. □

Remark 15.2. Of course, one could loosen condition (b) to be $-\infty$ by taking $-g(x)$.

15.2 Power series

Definition 15.3. A *power series* is defined given a sequence $\{a_n\}_{n \geqslant 0}$ by:

$$\sum_{n=0}^{\infty} a_n z^n$$

It is immaterial whether this sequence converges or diverges for a particular z in this definition, though this is a question we will ask later.

A quick application of the root test gives:

Proposition 15.4. *Given a power series*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

put

$$\alpha = \limsup_{n \to \infty} \sqrt[n]{|a_n|}$$

and

$$R = \frac{1}{\alpha}$$

Then, $f(z)$ converges for $|z| < R$ and diverges for $|z| > R$. This R is called the radius of convergence.

Proposition 15.5. *Given a power series*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

put

$$\alpha = \limsup_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$$

Then the radius of convergence R has $R \geqslant \frac{1}{\alpha}$.

Example 15.6. We compute the radius of convergence:
(a) $\sum n^n z^n$ has $R = 0$
(b) $\sum \frac{z^n}{n!}$ has $R = \infty$ (use ratio test instead)
(c) $\sum z^n$ has $R = 1$. If $|z| = 1$ then the series diverges.
(d) $\sum \frac{z^n}{n}$ has $R = 1$. It diverges for $z = 1$ and converges for $z = -1$.
(e) $\sum \frac{z^n}{n^2}$ has $R = 1$ and converges for $|z| = 1$.

15.3 Taylor series

Definition 15.7. Let f be a function for which all higher derivatives exist at a. Then, define the *Taylor series* of f at a to be the formal power series:

$$T_f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Theorem 15.8 (Taylor's theorem)**.** *Let $f : [a,b] \to \mathbb{R}$, $f^{(n-1)}$ is continuous on $[a,b]$ (and therefore all lower derivatives are continuous), and $f^{(n)}$ exists on (a,b). Let $\alpha, \beta \in [a,b]$ be distinct points and define*

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t-\alpha)^k$$

Then there exists a point c between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(c)}{n!}(\beta-\alpha)^n.$$

Remark 15.9. Let $n = 1$. Then $P(t) = f(\alpha)$ and the statement is that there is $c \in (\alpha, \beta)$ such that

$$f(\beta) - f(\alpha) = f'(c)(\beta - \alpha)$$

i.e. the mean value theorem. Thus one can think of this as a "higher order mean value theorem."

Proof. Without loss of generality let $\alpha < \beta$. Choose M such that

$$f(\beta) = P(\beta) + M(\beta-\alpha)^n$$

and let

$$g(x) = f(x) - P(x) - M(t-\alpha)^n$$

so that $g(\beta) = 0$. We want to show that $n!M = f^{(n)}(x)$ for some $x \in (\alpha, \beta)$. One finds that

$$g^{(n)} = f^{(n)}(x) - n!M$$

so we've reduced our claim to showing that $g^{(n)}(x) = 0$ for some $x \in (\alpha, \beta)$.

Note that $g(\alpha) = 0$. By Rolle's theorem (here we need continuity of the derivatives at the endpoints) there is some $c_1 \in (\alpha, \beta)$ such that $g'(c_1) = 0$. Notice that also, $g'(\alpha) = 0$¡ and in fact, all higher derivatives up to $f^{(n-1)}(\alpha) = 0$. Thus we can repeat the argument for g' and (α, c_1), and then for g'' and (α, c_2), and at the end we find $c_n = c$. □

Remark 15.10. Note that Taylor's theorem doesn't say anything about convergence of the Taylor series. Consider the function:

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & x > 0 \\ 0 & x \leqslant 0 \end{cases}$$

All higher derivatives of this function exist and are continuous everywhere. However, its Taylor series at 0 is $T_f(x) = 0$.

16 Lecture 16

16.1 The Riemann-Stieltjes integral

We will have to introduce a lot of notation in this section. **For the entirety of this section $f : [a, b] \to \mathbb{R}$ will refer to a bounded function, and $\alpha : [a, b] \to \mathbb{R}$ a monotonically increasing weight function.** We will treat other cases later.

Definition 16.1. Let $[a, b]$ be an interval. Let $\alpha : [a, b] \to \mathbb{R}$ be a monotonically increasing weight function. If one would prefer to simpliy the exposition on a first reading, one can just fix $\alpha(x) = x$.

A *partition* P of $[a, b]$ can be given by a finite set of points $a = x_0 \leqslant x_1 \leqslant \cdots \leqslant x_n = b$ which we think of as endpoints of subintervals of $[a, b]$, i.e.

$$[a, b] = [x_0, x_1] \cup [x_1, x_2] \cup \cdots \cup [x_{n-1}, x_n]$$

Note that this is not a partition in the sense of equivalence relations since the intervals are not disjoint. Write:

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$$

i.e. the length of $[x_{i-1}, x_i]$. **Unfortunately one does have to remember the convention that Δx_i refers to the interval where x_i is the upper endpoint.**

Let

$$M_i = \sup\{f(x) \mid x \in [x_{i-1}, x_i]\}$$

$$m_i = \inf\{f(x) \mid x \in [x_{i-1}, x_i]\}$$

Note that both these exist since f is bounded.

Define the *upper and lower Darboux sums*:

$$U(P, f, \alpha) = \sum_{i=1}^{n} M_i \delta\alpha_i$$

$$L(P, f, \alpha) = \sum_{i=1}^{n} m_i \delta\alpha_i$$

Define the *upper and lower Darboux integrals*:

$$U(f, \alpha) = \inf\{U(P, f, \alpha) \mid \text{partitions } P\}$$

$$L(f, \alpha) = \sup\{L(P, f, \alpha) \mid \text{partitions } P\}$$

where we range over all partitions P.

If $U(f) = L(f)$ then we say f is *Riemann-integrable* and define the *Riemann-Stiljes integral*:

$$\int_a^b f \, d\alpha$$

Definition 16.2. Let P, Q be two partitions, given by points $x_0, \ldots, x_n$ and $y_0, \ldots, y_m$. We say that P is a

refinement of Q if $\{y_0, \ldots, y_m\} \subset \{x_0, \ldots, x_n\}$. Note that we forget about the "multiplicities" here, i.e. $P = (0, 1, 2, 3)$ is a refinement of $(0, 1, 1, 3)$. Further, any two partitions P and Q have a *common refinement* $P \cup Q$. The common refinement is unique only up to repetition of one of the points x_i.

Proposition 16.3. *If P is a refinement of Q, then*

$$U(P, f, \alpha) \leqslant U(Q, f, \alpha)$$

$$L(Q, f, \alpha) \leqslant L(P, f, \alpha)$$

Proof. We will only prove the first statement. It suffices to prove the case where P has only one more point than Q. If that point is already a point of Q there is nothing to prove, since $\delta\alpha_i = 0$ for that term, so suppose that the extra point of Q is $x_i \in (x_{i-1}, x_{i+1})$. Then,

$$U(Q, f, \alpha) - U(P, f, \alpha) = \sup\{f([x_{i-1}, x_{i+1}])\}(\alpha_{i+1} - \alpha_{i-1}) - \sup\{f([x_{i-1}, x_i])\}(\alpha_i - \alpha_{i-1}) - \sup\{f([x_i, x_{i+1}])\}(\alpha_{i+1} - \alpha_i)$$

But notice that $L := \sup f([x_{i-1}, x_{i+1}]) \geqslant \sup f(I)$ for any subinterval I,

$$U(Q, f, \alpha) - U(P, f, \alpha) \geqslant L(\alpha_{i+1} - \alpha_{i-1} - \alpha_i + \alpha_{i+1} - \alpha_{i+1} + \alpha_i) = 2L(\alpha_{i+1} - \alpha_{i-1}) \geqslant 0$$

□

Proposition 16.4. $U(f, \alpha) \geqslant L(f, \alpha)$

Proof. It is clear that for any particular P, one has $U(P, f, \alpha) \geqslant L(P, f, \alpha)$. Further, for any P_1 and P_2 with common refinement Q, one has:

$$L(P_1, f, \alpha) \leqslant L(Q, f, \alpha) \leqslant U(Q, f, \alpha) \leqslant U(P_2, f, \alpha)$$

Taking the sup over P_1 and the inf over P_2 gives:

$$L(f, \alpha) \leqslant U(f, \alpha)$$

□

The following is a sufficient condition for integrability.

Proposition 16.5. *f is Riemann integrable (with respect to α) if and only if for every $\epsilon > 0$ there is a partition P such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

Proof. Suppose the condition holds. Note that

$$0 \leqslant U(f, \alpha) - L(f, \alpha) \leqslant U(P, f, \alpha) - L(P, f, \alpha)$$

for any P, and the result follows. Conversely, if f is Riemann-integrable, then there exists a P such that

$$U(P, f, \alpha) - U(f, \alpha) < \frac{\epsilon}{2}$$

and a Q such that

$$L(f, \alpha) - L(P, f, \alpha) < \frac{\epsilon}{2}$$

and thus their common refinement has

$$U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

□

Example 16.6 ($\alpha(x) \neq x$)**.** We can consider the case where $\alpha(x)$ is a "step" function, i.e. take the floor function:

$$\alpha(x) = \lfloor x \rfloor$$

Now, what is $\int_a^b f\, d\alpha$, if it exists? Note that the only intervals which contribute to the sums are those which contain an integer, since otherwise one would have $\Delta\alpha_i = 0$. Further, we can refine any partition so that each interval contains at most one integer. Then as long as f is continuous at each integer, one has that (say $a, b \in \mathbb{Z}$):

$$\int_a^b f\, d\alpha = \sum_{i=a}^{b} f(i).$$

Example 16.7. As for another example, let's take $\alpha(x) = 2x$. Then one can check that:

$$\int_a^b f\, d\alpha = 2\int_a^b f\, dx$$

if the integral exists at all. We will see more examples later.

16.2 Some Riemann-integrable functions

Proposition 16.8. *If f is monotonic and α is continuous, then f is Riemann-integrable.*

Proof. Suppose that f is monotonically increasing. Then, in the Darboux sum, for the subinterval $[x_{i-1}, x_i]$, one has that $M_i = f(x_i)$ and $m_i = f(x_{i-1})$. Take P to be a partition where $\Delta\alpha_i = \frac{\alpha(b)-\alpha(a)}{n}$. Then,

$$U(P, f, \alpha) - L(P, f, \alpha) = \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^{n} (f(x_i) - f(x_{i-1})) = \frac{(\alpha(b) - \alpha(a))(f(b) - f(a)}{n}$$

For large n this becomes arbitrarily small. □

Proposition 16.9. *If f is continuous then f is Riemann-integrable for any α.*

Proof. Given $\epsilon > 0$, choose some $\eta > 0$ such that

$$(\alpha(b) - \alpha(a))\eta < \epsilon$$

Since f is continuous on $[a, b]$ and $[a, b]$ is compact, it is uniformly continuous, so there exists some $\delta > 0$ such that

$$|f(x) - f(y)| < \eta$$

for $|x - y| < \delta$. Now, take P to be a partition where $x_{i+1} - x_i < \delta$, so that

$$M_i - m_i \leqslant \eta$$

and so

$$U(P, f, \alpha) - L(P, f, \alpha) \leqslant \eta \sum \Delta\alpha = \eta(\alpha(b) - \alpha(a)) < \epsilon$$

□

Proposition 16.10. *Suppose f has finitely many points of discontinuity, and α is continuous at every discontinuous point of f. Then f is Riemann-integrable.*

Proof. The idea is as follows: where f is continuous we can bound $M_i - m_i$. Where α is continuous we can bound $\alpha(x_i) - \alpha(x_{i-1})$. We formalize this as follows. Let $p_1, \ldots, p_k$ be the points of discontinuity for f, and find intervals $[c_i, d_i]$ such that p_i is the only point of discontinuity in it and such that $\sum_i \alpha(d_i) - \alpha(c_i) < \epsilon$. Let M be an upper bound on f on $[a, b]$. Then one has that the Darboux sum on this part is less than

$$\sum (M_i - m_i)(\alpha(d_i) - \alpha(c_i)) \leqslant 2M \sum \alpha(d_i) - \alpha(c_i) = 2M\epsilon$$

Let $K = \overline{[a,b] - \bigcup [c_i, d_i]}$. This is compact, so f is uniformly continuous on this interval, so find a δ such that $|f(x) - f(y)| < \epsilon$ if $|x - y| < \delta$. Now, there is a partition P such that
(a) c_i and d_i are in P
(b) no point in (c_i, d_i) is in P
(c) every interval not $[c_i, d_i]$ has length less than δ

Then, one has that, following the proof of the previous proposition,

$$U(P, f, \alpha) - L(P, f, \alpha) \leqslant (\alpha(b) - \alpha(a))\epsilon \leqslant 2M\epsilon$$

□

Proposition 16.11. *Let f be Riemann-integrable, and $f([a,b]) \subset I$ for some closed interval I. Suppose that $g : I \to \mathbb{R}$ is continuous. Then $h := g \circ f$ is Riemann-integrable, with respect to any α.*

Proof. Fix an $\epsilon > 0$. Now, consider the sum we want to make small for some partition P (i.e. smaller than ϵ):

$$U(P, h, \alpha) - L(P, h, \alpha) = \sum (M_i^h - m_i^h)(\alpha(x_i) - \alpha(x_{i-1}))$$

where the M_i^h, m_i^h correspond to $h = g \circ f$. Let M_i^f and m_i^f be the corresponding numbers for f.

First, we use uniform continuity of g. There is positive $\delta > 0$ such that $|g(x) - g(y)| < \epsilon$ if $|x - y| \leqslant \delta$. Then, if $M_i^f - m_i^f < \delta$, i.e. $|f(x) - f(y)| < \delta$ for $x, y \in [x_{i-1}, x_i]$, one has that $M_i^h - m_i^h < \epsilon$.

However, if $M_i^f - m_i^f \geqslant \delta$, since f is Riemann-integrable, the intervals on which $M_i^f - m_i^f \geqslant \delta$ have to be "short enough." We can choose a partition $P = \{x_0, \ldots, x_n\}$ such that $U(P, f, \alpha) - L(P, f, \alpha) < \delta^2$. Then we have

$$\delta \sum \Delta\alpha_i \leqslant \sum (M_i^f - m_i^f)\Delta\alpha < \delta^2$$

$$\sum \Delta\alpha_i < \delta$$

where the sum is only taken over intervals where $M_i^f - m_i^f \geqslant \delta$. Note that this says that the intervals on which $M_i^f - m_i^f \geqslant \delta$ must be of total weighted length less than δ. Let K be an upper bound on $|g|$, so that $M_i^h - m_i^h \leqslant 2K$. Then,

$$\sum (M_i^h - m_i^h)\Delta\alpha_i < 2K\delta.$$

Putting these two cases together, and taking δ such that $\delta < \epsilon$, one has that

$$U(P, h, \alpha) - L(P, h, \alpha) \leqslant \epsilon(\alpha(b) - \alpha(a)) + 2K\delta < \epsilon(\alpha(b) - \alpha(a) + 2K)$$

□

17 Lecture 17

17.1 Properties of the integral

Proposition 17.1 (Linearity properties)**.** *Let f, g be Riemann integrable and let $c \in \mathbb{R}$, and α continuous at c. Then the following are also Riemann-integrable and have values:*

$$\int_a^b cf \, d\alpha = \int_a^b cf \, d\alpha$$

$$\int_a^b f + g \, d\alpha = \int_a^b f \, d\alpha + \int_a^b g \, d\alpha$$

Proof. These follow directly from the definition. Let M_i and m_i correspond to f and M_i' and m_i' correspond to g.

Then one has that for any P,

$$U(P, f+g, \alpha) \leqslant \sum (M_i + M_i')\Delta\alpha_i = U(P, f, \alpha) + U(P, g, \alpha)$$

The inequality is because the maximum of f and g may not occur at the same point. Simialrly,

$$L(P, f+g, \alpha) \geqslant L(P, f, \alpha) + L(P, g, \alpha)$$

and since $U(f, \alpha) = L(f, \alpha)$ and $U(g, \alpha) = L(g, \alpha)$, one has that $U(f+g, \alpha) = L(f+g, \alpha)$. □

Proposition 17.2 (Bounds of integration)**.** *If f is Riemann-integrable and $c \in [a, b]$, then the following as also Riemann-integrable and one has:*

$$\int_a^c f\, d\alpha + \int_c^b f\, d\alpha = \int_a^b f\, d\alpha$$

Proof. Any partition of $[a, b]$ can be refined to include the point c. The rest is left as an exercise. □

Proposition 17.3 (Comparison)**.** *Let f, g be Riemann integrable such that $f(x) \leqslant g(x)$ on $[a, b]$. Then*

$$\int_a^b f\, d\alpha \leqslant \int_a^b g\, d\alpha$$

Further, if $|f(x)| \leqslant M$, then

$$|\int_a^b f\, d\alpha| \leqslant M(\alpha(b) - \alpha(a))$$

Proof. Since $U(P, f, \alpha) \leqslant U(P, g, \alpha)$ for every partition P, taking inf over all P yields the result. For the second statement, then one has $U(P, f, \alpha) \leqslant M(\alpha(b) - \alpha(a)$ and similarly for $L(P, f, \alpha)$, for any P. □

Proposition 17.4 (Linearity in the weights)**.** *If f is Riemann integrable with respect to α_1 and also α_2, then the following are also Riemann-integrable:*

$$\int_a^b f\, d(\alpha_1 + \alpha_2) = \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2$$

$$\int_a^b f\, d(c\alpha) = c\int_a^b f\, d\alpha$$

Proof. This is immediate. □

Proposition 17.5 (Closure under products)**.** *If f is Riemann integrable, and g is Riemann integrable, then fg is Riemann integrable.*

Proof. By a previous proposition, if f is Riemann-integrable, then f^2 is. Then note that $(f+g)^2 = f^2 + 2fg + g^2$ and so $fg = \frac{1}{2}((f+g)^2 - f^2 - g^2)$, completing the proof. □

Proposition 17.6 (Triangle inequality)**.** *If f is Riemann integrable, then $|f|$ is, and*

$$|\int_a^b f\, d\alpha| \leqslant \int_a^b |f|\, d\alpha$$

Proof. For the first statement, note that since $g(x) = |x|$ is Riemann-integrable, if f is Riemann-integrable, then so is $|f|$. For the second statement, suppose $\int_a^b f\, d\alpha \geqslant 0$. Then since $f \leqslant |f|$ the statement follows. If $\int_a^b f\, d\alpha \leqslant 0$, then since $-f \leqslant |f|$, the statement follows. □

Proposition 17.7 (Intermediate value theorem for integrals or "average value" theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then there is an $x \in [a, b]$ such that*

$$f(x) = \frac{1}{b-a}\int_a^b f\, dx$$

Proof. Let M and m be the minimum and maximum values of f on $[a, b]$. Then,

$$m \leqslant \frac{1}{b-a} \int_a^b f \, dx \leqslant M$$

Now apply the intermediate value theorem. □

Proposition 17.8. *Suppose α increases monotonically, is everywhere differentiable, and α' is Riemann integrable. Then f is Riemann integrable with respect to α if and only if $f\alpha'$ is Riemann integrable with respect to x. In this case,*

$$\int_a^b f \, d\alpha = \int_a^b f\alpha' \, dx$$

Proof. It suffices to show the equality of the upper and lower integrals of $f \, d\alpha$ and $f\alpha' \, dx$. We will do the upper; the lower follows in the same way. Since α' is Riemann-integrable, for every $\epsilon > 0$ there is a P such that

$$U(P, \alpha', x) - L(P, \alpha', x) < \epsilon$$

By the mean value theorem, we can find points $c_i \in [x_{i-1}, x_i]$ such that

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(c_i)(x_i - x_{i-1}) = \alpha'(c_i)\Delta x_i$$

For any $t_i \in [x_{i-1}, x_i]$ one has:

$$\sum |\alpha'(t_i) - \alpha'(c_i)|\Delta x_i \leqslant \sum (M_i^{\alpha'} - m_i^{\alpha'})\Delta x_i < \epsilon$$

Let M be a bound for $|f|$. Then one has:

$$|\sum f(t_i)\Delta\alpha_i - \sum f(t_i)\alpha'(t_i)\Delta x_i| \leqslant \sum |f(t_i)||\Delta\alpha_i - \alpha'(t_i)\Delta x_i| \leqslant M \sum |\Delta\alpha_i - \alpha'(t_i)\Delta x_i|$$

$$\leqslant M(|\Delta\alpha_i - \alpha'(c_i)\Delta x_i| + |\alpha'(c_i)\Delta x_i - \alpha(t_i)\Delta x_i|) < M(\epsilon + 0) = M\epsilon$$

Lemma 17.9. *Let $c > 0$. Then $|x + y| < c$ if and only if $x < c + y$ and $y < c + x$.*

So,

$$\sum f(t_i)\Delta\alpha_i \leqslant \sum f(t_i)\alpha'(t_i)\Delta x_i + M\epsilon \leqslant U(P, f\alpha', x) + M\epsilon$$

$$\sum f(t_i)\alpha'(t_i)\Delta x_i \leqslant \sum f(t_i)\Delta\alpha_i + M\epsilon \leqslant U(P, f, \alpha) + M\epsilon$$

Since this is true for any choice of t_i, in particular $t + i = M_i$, one has:

$$U(P, f\alpha', x) \leqslant U(P, f, \alpha) + M\epsilon$$

$$U(P, f, \alpha) \leqslant U(P, f\alpha', x) + M\epsilon$$

So,

$$|U(P, f\alpha', x) - U(P, f, \alpha)| < M\epsilon$$

Since everything above is true for a refinement,

$$|U(f\alpha', x) - U(f, \alpha)| < M\epsilon$$

and so

$$U(F, \alpha', x) = U(f, \alpha)$$

□

Proposition 17.10. *Suppose $\phi : [A, B] \to [a, b]$ is strictly increasing and continuous. Suppose α is monotonically*

increasing on $[a,b]$ and $f:[a,b]\to\mathbb{R}$ is Riemann integrable with respect to α. Define by change of variables:

$$\beta(x)=\alpha(\phi(x))$$

$$g(x)=f(\phi(x))$$

Then g is Riemann integrable with respect to beta and

$$\int_A^B g\ d\beta=\int_a^b f\ d\alpha$$

Proof. Since ϕ is strictly increasing, a partiton of $[A,B]$ gives a partition of $[a,b]$ and vice versa. Let them be P and Q respectively. Then one finds that

$$U(P,g,beta)=U(Q,f,\alpha)$$

and likewise, and the result follows. □

Corollary 17.11. *Suppose ϕ' is Riemann integrable. Then,*

$$\int_a^b f(x)\ dx=\int_A^B f(\phi(y))\phi'(y)\ dy$$

Proof. Take $\alpha(x)=x$ and $\beta=\phi$ above. □

18 Lecture 18

18.1 The fundamental theorem of calculus

In this section we will take $\alpha(x)=x$.

Theorem 18.1 (The fundamental theorem of calculus I)**.** *Let f be Riemann integrable on $[a,b]$. Define a function F on $[a,b]$ by:*

$$F(x)=\int_a^x f(t)\ dt$$

Then F is continuous on $[a,b]$. Furthermore, f is continuous at a point x_0 if and only if F is differentiable at x_0 and:

$$F'(x_0)=f(x_0)$$

Proof. First we will show that F is uniformly continuous, and thus continuous. Let M be a bound on $|f|$. Then,

$$|F(y)-F(x)|=|\int_x^y f(t)\ dt|\leqslant M|x-y|$$

Thus, if $|x-y|<\frac{\epsilon}{M}$, then $|F(x)-F(y)|<\epsilon$, finishing the claim.

Now, suppose f is continuous at x_0. For any $\epsilon>0$ there is $\delta>0$ such that $|f(t)-f(x_0)|<\epsilon$ for $|t-x_0|<\delta$. Then for any $t\in(x_0-\delta,x_0+\delta)$ with $t\neq x_0$ and by the intermediate value theorem for integrals, there is a $c\in(t,s)$ such that

$$\frac{F(t)-F(x_0)}{t-x_0}=\frac{1}{t-x_0}\int_s^t f(x)\ dx=f(c)$$

so

$$|\frac{F(t)-F(x_0)}{t-x_0}-f(x_0)|=|f(c)-f(x_0)|<\epsilon$$

Thus one has that

$$F'(x_0)=\lim_{t\to x_0}\frac{F(t)-F(x_0)}{t-x_0}=f(x_0)$$

as desired. One obtains the converse by reversing these arguments. □

Theorem 18.2 (The fundamental theorem of calculus II)**.** *If f is Riemann integrable on $[a, b]$ which has an* antiderivative*, i.e. a function F such that $f' = f$, then*

$$\int_a^b f(x)\, dx = F(b) - F(a)$$

for any antiderivative F.

Proof. Fix $\epsilon > 0$. Let P be a partition such that $U(P, f) - L(P, f) < \epsilon$. Applying the mean value theorem to each interval, there is some $t_i \in [x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = F'(t_i)\Delta x_i = f(t_i)\Delta x_i$$

And so

$$\sum f(t_i)\Delta x_i = F(b) - F(a)$$

Since both the above sum and $\int_a^b f(x)$ is between $L(P, f)$ and $U(P, f)$, one has that

$$|F(b) - F(a) - \int_a^b f(x)\, dx| < \epsilon$$

□

Corollary 18.3 (Substitution)**.** *Let u be a differentiable function on (a, b) such that u' is continuous, and I be an open interval such that $u([a, b]) \subset I$. If f is continuous on I, then $f \circ u$ is continuous on J and*

$$\int_a^b f(u(x))u'(x)\, dx = \int_{u(a)}^{u(b)} f(u)\, du$$

Proof. Let F be as in the fundamental theorem. Then if $G = F \circ u$ then by the chain rule one has that $G'(x) = F(u(x))u'(x)$. Now one has

$$\int_a^b f(u(x))u'(x)\, dx = \int_a^b G'(x)\, dx = G(b) - G(a)F(u(b)) - F(u(a)) = \int_{u(a)}^{u(b)} f(u)\, du$$

□

Corollary 18.4 (Integration by parts)**.** *Suppose F and G are differentiable functions on $[a, b]$ and their derivatives f, g are Riemann integrable. Then*

$$\int_a^b F(x)g(x)\, dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\, dx$$

Proof. Put $H(x) = F(x)G(x)$. Then $H'(x) = F(x)g(x) + f(x)G(x)$, and apply the second fundamental theorem. □

19 Lecture 19

19.1 Things that aren't true

First, I want to introduce the most immediate version of convergence one can ask for of a function.

Definition 19.1. Let $f_n : I \to \mathbb{R}$ be a sequence of functions. Then one says they converge *pointwise* to a function $f : J \to \mathbb{R}$ (for $J \subset I$) if

$$\lim_{n\to\infty} f_n(x) = f(x)$$

for every fixed $x \in \mathbb{J}$.

Example 19.2 (Interchanging limits for sequences)**.** Define

$$s_{m,n} = \frac{m}{m+n}$$

Then,

$$\lim_{m\to\infty} \lim_{n\to\infty} s_{m,n} = \lim_{m\to\infty} 0 = 0$$

but

$$\lim_{n\to\infty} \lim_{m\to\infty} s_{m,n} = \lim_{n\to\infty} 1 = 1$$

Example 19.3 (Limit of continuous functions is not continuous)**.** Define

$$f_n(x) = \frac{x^2}{(1+x^2)^n}$$

and the partial sums

$$F_n(x) = \sum_{k=1}^{n} f_n(x)$$

Then,

$$\lim_{n\to\infty} F_n(x) = \begin{cases} 0 & x = 0 \\ 1 + x^2 & x \neq 0 \end{cases}$$

since for fixed x, the f_n give a geometric sequence with first term x^2 and ratio $\frac{1}{1+x^2}$, and so

$$x^2 \frac{1}{1 - \frac{1}{1+x^2}} = 1 + x^2$$

This function is not continuous.

Example 19.4 (Limit of derivatives is not derivative of limit)**.** Define

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$$

and define

$$f(x) = \lim_{n\to\infty} f_n(x) = 0$$

Then we have that

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = 0$$

but

$$\lim_{n\to\infty} \frac{d}{dx} f_n(x) = \lim_{n\to\infty} \sqrt{n}\cos(nx) = \infty$$

Example 19.5 (Limit of integrals is not integral of limits)**.** Define

$$f_n(x) = n^2 x(1-x^2)^n$$

Then one has that

$$\int_0^1 f_n(x)\, dx = n^2 \int_0^1 x(1-x^2)^n\, dx = n^2 (\frac{1}{2(n+1)}(1-x^2)^{n+1})|_0^1 = \frac{n^2}{2n-2}$$

So,

$$\lim_{n\to\infty} \int_0^1 f_n(x) = \infty$$

However,

$$\lim_{n\to\infty} n^2 x(1-x^2)^n = 0$$

for $x \in [0,1]$, and so

$$\int_0^1 \lim_{n\to\infty} f_n(x) = 0$$

19.2 Uniform convergence

Definition 19.6. Let f_n be a sequence of functions which converges pointwise to f. Then we say f_n converges *uniformly* to f on E if for every $\epsilon > 0$ there is an N such that for $n > N$, one has

$$|f_n(x) - f(x)| < \epsilon$$

for every $x \in E$.

Example 19.7. Take $E = (-1,1)$ and define

$$f_n(x) = (1-|x|)^n$$

One can check that this converges pointwise to

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

However, it does not converge uniformly. When $x \neq 0$ one has that

$$|f(x) - f_n(x)| = (1-|x|)^n$$

However, for $\epsilon = \frac{1}{2}$ and any n, one can find $x \neq 0$ such that

$$x^2 < 1 - \sqrt[n]{\frac{1}{2}}$$

Example 19.8. Define

$$f_n(x) = \frac{1}{n}\sin(nx)$$

One can check that this converges to 0 pointwise. It converges uniformly.

Example 19.9. Take $E = [0,1]$ and

$$f_n(x) = x^n$$

This converges to

$$f(x) = \begin{cases} 0 & x \neq 1 \\ 1 & x = 1 \end{cases}$$

This does not converge uniformly.

Each of the previous examples do not converge uniformly. I encourage you to do the following as exercises and check against these notes.

Example 19.10. Take $E = \mathbb{N}$ and define,

$$s_m(n) = \frac{m}{m+n}$$

This does not converge uniformly in E. For fixed $n \in E$, the limit of the sequence is 1. However,

$$|\frac{m}{m+n} - 1| = |\frac{n}{m+n}|$$

For, say, $\epsilon = \frac{1}{2}$ and any m, there is some n such that

$$|\frac{n}{m+n}| > \epsilon$$

for example, $m < n$.

Further, one can do this the other, i.e. take

$$s_n(m) = \frac{m}{m+n}$$

This also does not converge uniformly in E. For any fixed $m \in E$ the limit of the sequence is 0. However, for $\epsilon = \frac{1}{2}$ and any n, there is an m such that

$$\frac{m}{m+n} < \frac{1}{2}$$

for example, choose $n > m$.

Example 19.11. Define

$$f_n(x) = \frac{x^2}{(1+x^2)^n}$$

and the partial sums

$$F_n(x) = \sum_{k=1}^{n} f_n(x)$$

so that

$$F(x) = \lim_{n\to\infty} F_n(x) = \begin{cases} 0 & x = 0 \\ 1 + x^2 & x \neq 0 \end{cases}$$

Then one has that

$$|F(x) - F_n(x)| = \begin{cases} 0 & x = 0 \\ (1+x^2)^{1-n} & x \neq 0 \end{cases}$$

The sequence cannot be uniformly continuous, since for say $\epsilon = \frac{1}{2}$ and any n

$$(1+x^2)^{1-n} > \frac{1}{2}$$

is equivalent to

$$2 > (1+x^2)^{n-1}$$

which can be achieved by taking x such that

$$x^2 < \sqrt[n-1]{2} - 1$$

which is possible since $\sqrt[n-1]{2} - 1 > 0$.

Example 19.12. Take $E = [0, 1]$ and define

$$f_n(x) = n^2 x(1-x^2)^n$$

so that

$$\lim_{n\to\infty} n^2 x(1-x^2)^n = 0$$

However, one has that

$$n^2 x(1-x^2)^n > 1$$

$$(1-x^2)^n > x(1-x^2)^n > \frac{1}{n^2}$$

$$x^2 < 1 - \sqrt[n]{\frac{1}{n^2}}$$

20 Lecture 20

20.1 Basic criteria for uniform convergence

Proposition 20.1 (Cauchy criterion)**.** *Let $E \subset \mathbb{R}$, and $f_n : E \to \mathbb{R}$. This sequence converges uniformly if and only if for every $\epsilon > 0$ there exists an integer N such that for $m, n \geqslant N$, one has*

$$|f_n(x) - f_m(x)| < \epsilon$$

Proof. Suppose f_n converges uniformly to f. Then for every ϵ there is an N such that for $n > N$ one has $|f_n(x) - f(x)| < \frac{\epsilon}{2}$. Then, by the triangle inequality,

$$|f_n(x) - f_m(x)| \leqslant |f_n(x) - f(x)| + |f(x) - f_n(x)| < \epsilon$$

for $n, m > N$.

For the converse, using the usual Cauchy criterion, we know that f_n converges pointwise to some function f. Let $\epsilon > 0$ be arbitrary, and suppose the condition above holds. Then,

$$|f_n(x) - f_m(x)| < \epsilon$$

for $n, m > N$. Taking $m \to \infty$, one has the required condition for uniform convergence. □

Proposition 20.2. *Suppose f_n converges pointwise to f on E. Then f_n converges uniformly on E if and only if*

$$\lim_{n\to\infty} \sup_{x\in E} |f_n(x) - f(x)| = 0.$$

Proof. This is a rewording of the definition. □

Proposition 20.3 (M-test)**.** *Suppose f_n is a sequence on E and suppose that we have bounds (for $x \in E$)*

$$|f_n(x)| \leqslant M_n$$

for all n. If $\sum M_n$ converges, then $\sum f_n$ converges uniformly.

Proof. If $\sum M_n$ converges, then for any $\epsilon > 0$ there is N such that if $m, n > N$ then

$$\sum_{i=n}^{m} M_i < \epsilon$$

Then, one has

$$|\sum_{i=n}^{m} f_i(x)| \leqslant \sum_{i=n}^{m} M_i < \epsilon$$

□

20.2 Uniform convergence and continuity

Proposition 20.4. *Let X be a metric space, and let $E \subset X$. Let $f_n, f : E \to \mathbb{R}$, and $f_n \to f$ uniformly. Let p be a limit point of E, and suppose that*

$$\lim_{x\to p} f_n(x) = A_n$$

Then $\lim A_n$ converges and

$$\lim_{x\to p} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \lim_{x\to p} f_n(x)$$

i.e.

$$\lim_{x\to p} f(x) = \lim_{n\to\infty} A_n$$

Proof. FIx $\epsilon > 0$. By uniform convergence, there is N such that for $n, m > N$ one has

$$|f_n(x) - f_m(x)| < \epsilon$$

for all $x \in E$. Taking $x \to p$ one has

$$|A_n - A_m| \leqslant \epsilon$$

and so the sequence A_n converges. Let its limit be A.

Now, we want to show that

$$\lim_{x \to p} f(x) = A$$

Note that

$$|f(x) - A| \leqslant |f(x) - f_n(x)| + |f_n(x) - A_n| + |A_n - A|$$

We can make $|f(x) - f_n(x)| < \epsilon$ for all $x \in E$ by uniorm continuity. We can also make $|A_n - A| < \epsilon$ by convergence. Finally we can make $|f_n(x) - A_n| < \epsilon$ in a neighborhood of p and for large n. Together, one has that

$$|f(x) - A| \leqslant 3\epsilon$$

for x in a neighborhood of p (note n plays no role now), proving the result. □

Corollary 20.5. *Let $f_n \to f$ uniformly, and f_n continuous. Then f is continuous.*

Proposition 20.6. *Let $E = [a, b]$. Let α be monotonically increasing. Suppose $f_n : E \to \mathbb{R}$ are integrable and $f_n \to f$ uniformly. Then f is integrable and*

$$\int_a^b f_n \, d\alpha = \lim_{n \to \infty} \int_a^b f_n \, d\alpha$$

Proof. Let

$$\epsilon_n = \sup_{x \in E} |f_n(x) - f(x)|$$

By uniform continuity $\epsilon_n \to 0$. Further one has that

$$f_n - \epsilon_n \leqslant f \leqslant f_n + \epsilon_n$$

and taking integrals

$$\int_a^b (f_n - \epsilon_n) \, d\alpha \leqslant L(f, \alpha) \leqslant U(f, \alpha) \leqslant \int_a^b (f + \epsilon_n) \, d\alpha$$

and thus

$$U(f, \alpha) - L(f, \alpha) \leqslant 2 \int_a^b \epsilon_n \, d\alpha = 2(\alpha(b) - \alpha(a))\epsilon_n$$

Taking the limit $n \to \infty$ one has that f is integrable. Further we have that

$$|\int_a^b f \, d\alpha - \int_a^b f_n \, d\alpha| \leqslant \epsilon_n(\alpha(b) - \alpha(a))$$

□

Corollary 20.7 (Integration of series)**.** *Let f_n be integrable and the series $f(x) = sum f_n(x)$ converge uniformly. Then, $\int_a^b f \, d\alpha = \sum \int_a^n f_n \, d\alpha$.*

21 Lecture 21

21.1 Uniform convergence and differentiation

Proposition 21.1. *Let $E = [a, b]$ and let f_n be differentiable on E. Suppose f_n converges pointwise, for some point $p \in E$. If f_n' converges uniformly on E, then f_n converges uniformly on E, and*

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

Proof. First, we will show that f converges uniformly on E. Fix $\epsilon > 0$ and choose N such that $m, n > N$ implies (by pointwise convergence)

$$f_n(x_0) - f_m(x_0)| < \frac{\epsilon}{2}$$

and also (by uniform convergence)

$$|f_n'(x) - f_m'(x)| < \frac{\epsilon}{2(b-a)}$$

for all $x \in E$. Now, apply the mean value theorem to $f_n - f_m$ at any two points x, y:

$$|f_n(x) - f_m(x) - f_n(y) + f_m(y)| \leqslant \frac{|x-y|\epsilon}{2(b-a)} \leqslant \frac{\epsilon}{2}$$

Then take $y = x_0$:

$$|f_n(x) - f_m(x)| \leqslant |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \leqslant \epsilon$$

and so f_n converges uniformly.

Let f be its limit, and take $p \in E$. Define

$$\phi_n(x) = \frac{f_n(x) - f_n(p)}{x - p}$$

$$\phi(x) = \frac{f(x) - f(p)}{x - p}$$

By definition of derivative, $\lim_{x\to p} \phi_n(x) = f_n'(x)$. Using the previous inequality, one has that

$$|\phi_n(x) - \phi_m(x)| \leqslant \frac{\epsilon}{2(b-a)}$$

for large n, m, so that ϕ_n converges uniformly for $x \neq p$, and it must converge pointwise (and thus uniformly) to ϕ. Thus, by interchanging limits, one has that

$$\lim_{x\to p} \phi(x) = \lim_{n\to\infty} f_n'(x)$$

and the result follows. □

21.2 An everywhere continuous but nowhere differentiable function

Example 21.2. Define $\phi(x) = |x|$ for $x \in [-1, 1]$, and define it for the rest of $\mathbb{R}$ periodically, i.e. so that $\phi(x + 2) = \phi(x)$. Define

$$f(x) = \sum_{n=0}^{\infty} (\frac{3}{4})^n \phi(4^n x)$$

This might help think of this function: each term $(\frac{3}{4})^n \phi(4^n)$ "squeezes" the function ϕ horizontally by a factor of $\frac{1}{4^n}$ and also squeezes it vertically by a factor of $(\frac{3}{4})^n$. One can draw ϕ as a spiky function, and in each summand in $f(x)$, the spikes (discontinuous points) get shorter and also closer together. They have to get shorter in order for the function to converge, and they have to get closer together in order for the derivative to fail to exist everywhere.

More formally, since ϕ is bounded, the M-test for uniform convergence shows that this series converges uniformly, and so f is continuous. To see that it is not differentiable anywhere, take $x \in \mathbb{R}$. Notice that non-differentiable points of ϕ occur at the integers, so for any k we want to choose a δ_k such that ϕ has no integer points in $[x, x+\delta_k]$. Explicitly, we choose

$$\delta_k = \pm\frac{1}{2}4^{-k}$$

Define

$$\gamma_n = \frac{\phi(4^n(x+\delta_m)) - \phi(4^n x)}{\delta_m}$$

when $n > m$, $\delta_m 4^n$ is an even integer, so by periodicity, $\gamma_n = 0$. Further, one has that for any s, t:

$$|\phi(s) - \phi(t)| \leqslant |s - t|$$

so for $n < m$ one has $|\gamma_n| < 4^n$, and for $n = m$ one has that $\gamma_m = 4^m$.

Then, take

$$\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m} (\frac{3}{4})^n \gamma_n\right| \geqslant 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1)$$

As $m \to \infty$ one has $\delta_m \to 0$. So that:

$$f'(x) \geqslant \frac{1}{2}(3^m + 1)$$

for every m, so f is not differentiable at x.

21.3 Differentiation and integration of power series

Proposition 21.3. *Suppose the series*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

has radius of convergence R. Then it converges uniformly on any $[-R+\epsilon, R-\epsilon]$ for any $\epsilon > 0$. Further, f is continuous and differentiable in $(-R, R)$, and one can differentiate term-by-term:

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

and has radius of convergence R. Further, one can integrate term-by-term:

$$\int_0^x f(t)\, dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

and has radius of convergence R.

Proof. One has that

$$|a_n x^n| \leqslant M_n := |a_n|(R-\epsilon)^n$$

and since $\sum M_n$ converges absolutely by the root test, the M-test tells us that the series converges absolutely. The rest of the facts follow from previous propositions. □

Proposition 21.4 (Abel's theorem)**.** *Take*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

Let R be the radius convergence, and suppose f converges at R. Then, f is continious at R, i.e.

$$\lim_{x \to R} f(x) = \sum_{n=0}^{\infty} a_n R^n.$$

The same is true for $-R$.

Proof. Without loss of generality assume $R = 1$ (otherwise, we can replace x with x/R). Define

$$S_n = a_0 + \cdots + a_n$$

We can rewrite

$$f(x) = (1-x)\sum_{n=0}^{\infty} S_n x^n$$

Let $S = \lim S_n$. For $\epsilon > 0$, there is N such that for $n > N$ one has $|S - S_n| < \frac{\epsilon}{2}$. Then,

$$|f(x) - S| = |(1-x)\sum(S_n - S)x^n| \leqslant (1-x)(\sum |S_n - S||x|^n + \frac{\epsilon}{2}) \leqslant \epsilon$$

So as near 1, one has that $(1-x) \to 0$ and since ϵ was arbitrary the result follows. □

22 Lecture 22

22.1 The Stone-Weierstrass theorem

Definition 22.1. A *family* of real-valued functions on a set E is a subset of the set of all functions from E to $\mathbb{R}$. A family $\mathcal{A}$ is said to be an *algebra* if it is closed under addition, multiplication, and scalar multiplication in $\mathbb{R}$. A family $\mathcal{A}$ is said to be *uniformly closed* if for any sequence f_n with $f_n \in \mathcal{A}$, such that $f_n \to f$ uniformly, one has that $f \in \mathcal{A}$. Further, given any family $\mathcal{A}$ one can take its *uniform closure*, i.e. the smallest set family containing $\mathcal{A}$ that is uniformly closed.

Example 22.2. For example, the family of polynomial functions on $E = \mathbb{R}$ is an algebra, since the addition and multiplication of polynomials are also polynomials. Also, the family of functions of the form $P(x)e^x$ form an algebra, where P is any polynomial. Neither of these are uniformly closed, as we will see later.

Proposition 22.3. *Let $\mathcal{B}$ be the uniform closure of $\mathcal{A}$. Then $\mathcal{B}$ consists of all functions which arise as a uniformly convergent limit of functions in $\mathcal{A}$.*

Definition 22.4. Let $\mathcal{A}$ be a family of functions on a set E. Then $\mathcal{A}$ *separates points* if for any $x_1, x_2 \in E$, there is a function $f \in \mathcal{A}$ such that $f(x_1) \neq f(x_2)$; and it *vanishes at no point* if for every $x \in E$ there is some $f \in \mathcal{A}$ such that $f(x) \neq 0$.

Example 22.5. The family of polynomials on $\mathbb{R}$ separates points: the function $f(x) = x$ separates all points. Further, it vanishes at no point, since the function $f(x) = 1$ is nowhere zero

Example 22.6. The family of even polynomials on $\mathbb{R}$ does not separate points, since $f(x) = -f(x)$ for every such f.

Theorem 22.7. *Let $\mathcal{A}$ be an algebra of real continuous functions on a compact set K. If $\mathcal{A}$ separates points and vanishes at no point, then the uniform closure $\mathcal{B}$ consists of all real continuous functions on K.*

Proof. This is a long proof. We will divide it into five lemmas.

Lemma 22.8 (Stone-Weierstrass theorem)**.** *Let $\mathcal{A}$ be an algebra of continuous functions $[a,b] \to \mathbb{R}$ which contains $f(x) = x$. Then, $\mathcal{B}$ contains every continuous function on $[a,b]$.*

Proof. First, note that we may as well assume that $[a,b] = [0,1]$, since one can take $\tilde{f}(x) = f(\frac{x-a}{b-a})$ which has domain $[0,1]$, find a sequence of polynomials $\tilde{P}_n$ converging to it, and take $P_n(x) = \tilde{P}_n((b-a)/x + a)$.

The idea is as follows: the Dirac delta "function" is a function $\delta_a(x)$ which is "defined" by:

$$\delta_a(x) = \begin{cases} \infty & x = a \\ 0 & x \neq 1 \end{cases}$$

It has the "property" that:

$$\int_{-\infty}^{\infty} \delta_a(x) f(x)\, dx = f(a)$$

I will disgress for a bit to justify this property. Since δ_a is not really a function, we want to express it as a limit (which does not actually converge) of functions of the form:

$$Q_n(x) = \begin{cases} \frac{n}{2} & x \in [a - \frac{1}{n}, a + \frac{1}{n}] \\ 0 & \text{else} \end{cases}$$

The constant is chosen so that

$$\int_{-\infty}^{\infty} Q_n\, dx = 1$$

for all n. Then, one can verify that, if F is the antideriative of f, then

$$\int_{-\infty}^{\infty} Q_n(x) f(x)\, dx = \frac{n}{2} \int_{a-\frac{1}{n}}^{a+\frac{1}{n}} f\, dx = \frac{F(a + \frac{1}{n}) - F(a - \frac{1}{n})}{\frac{1}{n} - (-\frac{1}{n})}$$

Taking the limit as $n \to \infty$, one obtains $F'(a) = f(a)$ as desired.

Further, convolution gives:

$$(f * \delta_a)(x) = \int_{-\infty}^{\infty} f(t)\delta_a(x - t)\, dt = f(x - a)$$

In particular, if $a = 0$:

$$(f * \delta_0)(x) = \int_{-\infty}^{\infty} f(t)\delta_0(x - t)\, dt = f(x)$$

Our strategy is as follows: the Dirac delta function is not really a function, so we will take a sequence of *polynomial* functions that "limit" to it. That sequence of functions is:

$$Q_n(x) = \begin{cases} c_n(1 - x^2)^n & x \in [-1, 1] \\ 0 & x \notin [-1, 1] \end{cases}$$

where c_n is chosen such that

$$\int_{-\infty}^{\infty} Q_n(x)\, dx = 1$$

And we will claim that

$$P_n(x) := \int_{-\infty}^{\infty} Q_n(x - t) f(t)\, dt$$

converge uniformly to f (where we extend f to $\mathbb{R}$ by zero).

First we need to show that $P_n(x)$ is actually a polynomial. To this end, notice that (since f is defined in $[0, 1]$)

$$P_n(x) = \int_0^1 Q_n(x - t) f(t)\, dt$$

Using integration by parts, and letting F be the antiderivative of f, one has that

$$P_n(x) = Q(x - t)F(t)|_0^1 - \int_0^1 \frac{d}{dt} Q_n(x - t) F(t)\, dt$$

The term $Q(x - t)F(t)|_a^b$ is a polynomial, and in the integral term the polynomial $\frac{d}{dt} Q_n(x - t)$ is of smaller degree than $Q_n(x - t)$, so by induction one has that $P_n(x)$ is a polynomial. Note that this relies on the fact that the truncated polynomial $Q(x - t)$ is a polynomial for $t \in [0, 1]$ and $x \in [0, 1]$

Next, we want to show that P_n converges uniformly to f. Let us try to work out the expression first. All

integrals below are over all of $\mathbb{R}$:

$$|P_n(x) - f(x)| = \left| \int Q_n(x-t)f(t) - Q_n(x-t)f(x)\, dt \right|$$

since $\int Q_n(t)\, dt = 1$. Let us do a change of variables:

$$\left| \int Q_n(s)(f(x-s) - f(x))\, ds \right|$$

In this expression we are mostly worried about the integral near 0, where Q_n becomes very large (recall that the Dirac delta function "takes on" the value ∞ at 0), and it won't be enough to just replace the expression $f(x-s)-f(x)$ with a constant upper bound.

Our first step is to evaluate c_n. We compute

$$\int_{-1}^{1} (1-x^2)^n\, dx = 2\int_0^1 (1-x^2)^n\, dx$$

At this point one wants to replace $(1-x^2)^n$ with $1-nx^2$. One could try to work out the binomial theorem explicitly, but a short cut is to take the function

$$(1-x^2)^n - (1-nx)$$

whose derivative is

$$2nx(1-(1-x^2)^{n-1})$$

which is positive on $(0,1)$, and since they are equal at 0, one has

$$(1-x^2)^n \geqslant 1-nx$$

on $[0,1]$. Thus,

$$\int_{-1}^{1} (1-x^2)^n\, dx \geqslant 2\int_0^1 1-nx\, dx \geqslant \int_0^{1/\sqrt{n}} 1-nx\, dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}}$$

so, $c_n < \sqrt{n}$. The choice of taking the upper limit of $\frac{1}{\sqrt{n}}$ is a bit arbitrary, but it makes the expression work out nicely.

Next, we will analyze the behavior of Q_n away from $[-\delta, \delta]$. Since the function $Q_n(x)$ is decreasing in $[0,1]$, one has that for any $\delta > 0$ and any $x \in [\delta, 1]$:

$$Q_n(x) \leqslant \sqrt{n}(1-\delta^2)^n$$

and thus Q_n converges to zero uniformly in $[\delta, 1]$. Further, let M be a bound on f.

Now, by uniform continuity f ($[a,b]$ is compact), for every $\epsilon > 0$ there is a $\delta > 0$ such that $|x-y| < 2\delta$ implies that $|f(x) - f(y)| < \epsilon$.

Given these, we can evaluate:

$$\left| \int Q_n(s)(f(x-s) - f(x))\, ds \right| \leqslant 2M \left(\int_{-1}^{-\delta} Q_n(t)\, dt + \int_{\delta}^{1} Q_n(t)\, dt \right) + \epsilon \int_{-\delta}^{\delta} Q_n(t)\, dt$$

$$\leqslant 4M\sqrt{n}(1-\delta^2)^n + \epsilon < 2\epsilon$$

for large n, and since $\int_{-\delta}^{\delta} Q_n(t)\, dt \leqslant 1$, and we are done.

□

Lemma 22.9. *If $f \in \mathcal{B}$, then $|f| \in \mathcal{B}$ (where $\mathcal{B}$ is any uniformly closed algebra).*

Proof. $f(K) \subset [a,b] \subset \mathbb{R}$. For any $\epsilon > 0$, there is a polynomial

$$P(x) = a_0 + a_1 x + \cdots + a_n x^n$$

such that:

$$|P(x) - |x|| < \epsilon$$

for any $x \in [a, b]$. Then, the function

$$P(f) = a_0 + a_1 f + \cdots + a_n f^n$$

is in $\mathcal{B}$ and further one has that

$$|P(f(x)) - |f(x)|| < \epsilon$$

so that $|f| \in \mathcal{B}$ by closedness. □

Lemma 22.10. *If $f \in \mathcal{B}$ and $g \in \mathcal{B}$, then $\max(f, g) \in \mathcal{B}$ and $\min(f, g) \in \mathcal{B}$ (where $\mathcal{B}$ is any uniformly closed algebra).*

Proof. This follows from the fact that:

$$\max(f, g) = \frac{f + g}{2} + \frac{|f - g|}{2}$$

$$\min(f, g) = \frac{f + g}{2} - \frac{|f - g|}{2}$$

□

Lemma 22.11. *Suppose that $\mathcal{A}$ is an algebra of functions on any set E which separates points and vanishes at no point. Let $x_1 \neq x_2 \in E$ and $c_1, c_2 \in \mathbb{R}$. Then $\mathcal{A}$ has a function such that $f(x_1) = c_1$ and $f(x_2) = c_2$.*

Proof. This is best done as an exercise, but choose a functions g_i which do not vanish at x_i and h be a function such that $h(x_1) \neq h(x_2)$. Then take

$$f(x) = \frac{c_1 g_1(x)(h(x) - h(x_2))}{g_1(x_1)(h(x_1) - h(x_2))} + \frac{c_2 g_2(x)(h(x) - h(x_1))}{g_2(x_2)(h(x_2) - h(x_1))}$$

the key observation being that $g_1(x)(h(x) - h(x_2))$ is nonzero at x_1 and zero at x_2 (we need h to separate x_1 and x_2). Also, notice that constant functions are not automatically in $\mathcal{A}$, so we need to introduce the functions g_i to multiply the constants by, i.e. we need g_i to not vanish at x_i. □

The strategy is as follows. Let f be the continuous function we want to approximate. We will find for each point $p \in K$ a function that is within ϵ of f on one side, i.e. no less than $f - \epsilon$. Using compactness we will choose finitely many of these and take the maximum of the functions. We will do the same again on the other side.

Lemma 22.12. *Let $\mathcal{A}$, $\mathcal{B}$ as in the theorem. Let f be continuous on K (where K is compact). Let $p \in K$ and $\epsilon > 0$. There exists a function $g_p \in \mathcal{B}$ such that $g_p(p) = f(p)$ and $g_p(x) > f(x) - \epsilon$.*

Proof. For every $q \in K$, we can find $h_q \in \mathcal{B}$ such that

$$h_q(p) = f(p), h_q(q) = f(q)$$

Define the open set

$$J_q := (f - h_q)^{-1}((-\infty, \epsilon)$$

The J_q as $q \in K$ form a cover, since $q \in J_q$. Since K is compact there is a finite subcover indexed by I, and define the function

$$g_p = \max(h_{y_i})_{i \in I}$$

□

We will now show that given $f : K \to \mathbb{R}$ continuous, and $\epsilon > 0$, there is a function $h \in \mathcal{B}$ such that for $x \in K$:

$$|h(x) - f(x)| < \epsilon$$

From the previous lemma, for any $p \in K$, we have g_p such that

$$g_p(x) > f(x) - \epsilon$$

Define

$$V_p := (g_p - f)^{-1}((-\infty, \epsilon))$$

i.e. points such that

$$g_p(x) < f(x) + \epsilon$$

By compactness, this open cover has a finite subcover, and take

$$h = \min(g_{x_i})_{i \in I}$$

This function h has the property that

$$h(x) > f(x) - \epsilon$$

$$h(x) < f(x) + \epsilon$$

for $x \in K$. $\square$